

Bayesian Logistic Regression for Predicting Diabetes Risk Using NHANES 2013–2014 Data: A Capstone Project on Bayesian Applications in Epidemiologic Modeling

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Introduction

Diabetes mellitus (DM) remains a major public health challenge, and identifying key risk factors—such as obesity, age, sex, and race/ethnicity—is essential for prevention and targeted intervention. Logistic regression is widely used to estimate associations between such factors and binary outcomes like diabetes diagnosis. However, classical maximum likelihood estimation (MLE) can produce unstable estimates in the presence of missing data, quasi-separation, or small samples. Bayesian logistic regression offers a robust alternative by integrating prior information, regularizing estimates, and quantifying uncertainty more transparently than frequentist approaches.

Bayesian hierarchical models, implemented via Markov Chain Monte Carlo (MCMC), have been successfully applied in predicting patient health status across diseases such as pneumonia, prostate cancer, and mental disorders (Zeger et al. 2020). By representing predictive uncertainty alongside point estimates, Bayesian inference offers a practical advantage in epidemiologic modeling where decisions hinge on probabilistic thresholds. Beyond stability, Bayesian methods support model checking, variable selection, and uncertainty quantification under missingness or imputation frameworks (Baldwin and Larson 2017; Kruschke and Liddell 2017).

Recent work has expanded Bayesian applications to disease diagnostics and health risk modeling. For instance, Bayesian approaches have been used to evaluate NHANES diagnostic data (Chatzimichail and Hatjimihail 2023), to model cardiovascular and metabolic risk (Liu et al. 2013), and to integrate multiple data modalities such as imaging and laboratory measures (Abdullah, Hassan, and Mustafa 2022). Moreover, multiple imputation combined with

Bayesian modeling generates robust estimates when data are missing at random (MAR) or not at random (MNAR) (Austin et al. 2021).

The broader Bayesian literature emphasizes the role of priors and model checking. Weakly informative priors, such as for coefficients, regularize estimation and reduce variance in small samples (Gelman et al. 2008; Vande Schoot et al. 2021). Tutorials using R packages like `brms` and `blavaan` illustrate how MCMC enables posterior inference and empirical Bayes analysis (Klauenberg et al. 2015).

Beyond standard generalized linear models, Bayesian nonparametric regression flexibly captures nonlinearity and zero inflation common in health data (Richardson and Hartman 2018). Bayesian Additive Regression Trees (BART) improve variable selection in mixed-type data (Luo et al. 2024), while state-space and dynamic Bayesian models incorporate time-varying biomarkers for longitudinal prediction (Momeni et al. 2021). Bayesian model averaging (BMA) further addresses model uncertainty by weighting across multiple specifications (Hoeting et al. 1999). Together, these approaches demonstrate the versatility and growing importance of Bayesian inference in clinical and epidemiologic modeling.

The objective of this project is to evaluate whether Bayesian inference provides more stable and interpretable estimates of diabetes risk than frequentist and imputation-based approaches, particularly when data complexity or separation challenges arise. Agreement across modeling frameworks supports the robustness of these associations and highlights the interpretability and uncertainty quantification advantages offered by Bayesian analysis in population health modeling (National Center for Health Statistics (NCHS) 2014).

Aims

The present study employs Bayesian logistic regression to predict diabetes status and examine the relationships between diabetes and key predictors, including body mass index (BMI), age (20 years), sex, and race. Using retrospective data from the 2013–2014 NHANES survey, the analysis accounts for the study’s complex sampling design, which involves stratification, clustering, and the oversampling of specific sub-populations rather than simple random sampling. The Bayesian framework is applied to address common analytical challenges such as missing data, complete case bias, and data separation, thereby improving the robustness and reliability of inference compared to traditional logistic regression methods.

Method

Bayesian Logistic Regression

The Bayesian framework integrates prior knowledge with observed data to generate posterior distributions, allowing parameters to be interpreted directly in probabilistic terms.

Unlike traditional frequentist approaches that yield single-point estimates and p-values, Bayesian methods represent parameters as random variables with full

probability distributions.

This provides greater flexibility, incorporates parameter uncertainty, and produces credible intervals that directly quantify the probability that a parameter lies within a given range.

Model Structure

Bayesian logistic regression models the log-odds of a binary outcome as a linear combination of predictor variables. For participant i , the model is expressed as:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} \quad (1)$$

where p_i represents the probability of the event of interest for participant i , β_0 represents the intercept corresponding to the log-odds of the outcome when all predictors are equal to zero, and β_k represents the effect of predictor X_k on the log-odds of the outcome while holding all other predictors constant.

In the Bayesian framework, model parameters $(\beta_0, \beta_1, \dots, \beta_k)$ are considered random variables rather than fixed unknown quantities. Prior distributions are assigned to these parameters to incorporate existing knowledge or plausible parameter ranges before observing the data. After the observed data are incorporated, prior distributions are updated to obtain posterior distributions according to Bayes' theorem. ??:

$$P(\theta|Y) = \frac{P(Y|\theta)P(\theta)}{P(Y)} \quad (2)$$

where:

- $P(\theta|Y)$ represents the posterior distribution of the model parameters given the observed data.
- $P(Y|\theta)$ represents the likelihood function, which describes the probability of observing the data conditional on the model parameters.
- $P(\theta)$ represents the prior distribution, which reflects information or assumptions about the parameters before observing the data.
- $P(Y)$ represents the marginal probability of the observed data and serves as a normalizing constant.

This Bayesian formulation allows uncertainty to propagate naturally through the model by generating posterior distributions for each regression coefficient. Unlike frequentist point estimates, posterior distributions provide direct probabilistic interpretations of parameter uncertainty and allow estimation of credible intervals for regression effects.

Prior Specification

Weakly informative priors were assigned to model parameters to provide regularization while avoiding strong assumptions about the magnitude or direction of predictor effects. These priors allow the observed data to primarily influence parameter estimation while reducing the risk of unstable estimates caused by sparse data, multicollinearity, or separation issues.

Regression coefficients were assigned normal prior distributions:

$$\beta_j \sim \text{Normal}(0, 2.5^2) \quad (3)$$

where β_j represents the regression coefficient for predictor j . This weakly informative prior centers coefficients around zero while allowing substantial variation in plausible effect sizes. Such priors have been recommended to improve model stability and reduce variance in Bayesian regression models (??).

The intercept parameter was assigned a Student's t -distribution prior:

$$\beta_0 \sim \text{Student-}t(3, 0, 10) \quad (4)$$

where the distribution includes 3 degrees of freedom, a mean of 0, and a scale parameter of 10. The heavy-tailed nature of the Student's t distribution allows occasional large values while maintaining weak regularization. The mean of zero avoids imposing directional assumptions, and the broad scale allows flexibility in estimating baseline diabetes risk.(?).

These prior specifications provide a balance between incorporating prior knowledge and allowing the NHANES 2013–2014 data to determine posterior estimates. Weakly informative priors improve parameter stability, particularly in situations involving limited sample size, correlated predictors, or extreme observations.

Posterior Predictions

Posterior distributions of regression coefficients were used to estimate the probability of the outcome for given predictor values. This allows statements such as: > Given the predictors, the probability of the outcome lies between X% and Y%.

Posterior predictions account for two key sources of uncertainty:

1. **Parameter Uncertainty:** Variability in estimated model coefficients.
2. **Predictive Uncertainty:** Variability in possible future outcomes given those parameters.

In Bayesian analysis, all unknown quantities—coefficients, means, variances, or probabilities—are treated as random variables described by their posterior distributions.

Model Evaluation and Diagnostics

Model performance, convergence, and adequacy were evaluated using standard Bayesian diagnostic approaches. Posterior sampling was performed using Markov Chain Monte Carlo (MCMC) methods implemented through the No-U-Turn Sampler (NUTS), an efficient variant of Hamiltonian Monte Carlo (HMC) (?). Multiple independent chains were generated with adequate warm-up iterations to assess convergence and ensure reliable posterior estimation.

The following diagnostic measures were evaluated:

- **Posterior Sampling:** Posterior distributions of model parameters were estimated using MCMC sampling. Four independent chains were used with sufficient warm-up iterations to achieve stable sampling from the posterior distribution.
- **Convergence Diagnostics:** The potential scale reduction factor ($\hat{R}$) and effective sample size (ESS) were used to evaluate convergence and sampling efficiency. Values of $\hat{R}$ close to 1.0 indicate adequate convergence between chains, while larger ESS values indicate more reliable posterior estimates.
- **Autocorrelation Assessment:** Autocorrelation among successive MCMC draws was examined to evaluate sampling independence and ensure that posterior samples provided adequate information about the parameter distributions.
- **Posterior Predictive Checks (PPCs):** Posterior predictive checks were conducted by comparing simulated outcomes generated from the posterior distribution with the observed data. These assessments evaluated whether the model adequately reproduced important characteristics of the observed diabetes outcome distribution.
- **Bayesian R^2 :** Bayesian R^2 was calculated to quantify the proportion of variance in the outcome explained by the predictors while incorporating uncertainty from the posterior distribution.

Together, these diagnostic procedures provided an assessment of model convergence, predictive adequacy, and the reliability of Bayesian inference for diabetes risk estimation.

Advantages of Bayesian Logistic Regression

- **Uncertainty Quantification:** Produces full posterior distributions instead of single estimates.
- **Credible Intervals:** Provide the range within which a parameter lies with a specified probability (e.g., 95%).
- **Flexible Priors:** Allow integration of expert knowledge or findings from prior studies.

- **Probabilistic Predictions:** Posterior predictive distributions yield direct probabilities for new or future observations.
- **Model Evaluation:** PPCs assess how well simulated outcomes reproduce observed data.

Analysis and Results

Data Preparation

This study used publicly available 2013–2014 NHANES data published by the CDC’s National Center for Health Statistics (National Center for Health Statistics (NCHS) 2014). Three component files were utilized: DEMO_H (demographics), BMX_H (body measures), and DIQ_H (diabetes questionnaire). Each file was imported in .XPT format using the haven package in **R**, and merged using the unique participant identifier SEQN to create a single adult analytic dataset (age ≥ 20 years).

All variables were coerced to consistent numeric or factor types prior to merging to ensure atomic columns suitable for survey-weighted analysis and modeling. The use of SEQN preserved respondent integrity across datasets and ensured accurate record linkage. This preprocessing step standardized variable formats and minimized inconsistencies between files.

Data wrangling, cleaning, and merging were performed in **R** using a combination of base functions and tidyverse packages. Bayesian logistic regression modeling was later implemented using the brms interface to **Stan**, allowing probabilistic inference within a reproducible workflow that accommodated the NHANES complex survey design and missing data considerations.

Data Import and Merging

Code

```
merged_data <- readRDS("data/merged_2013_2014.rds")
merged_n <- nrow(merged_data)
```

The merged dataset contains 10,175 participants. It integrates demographic, examination, and diabetes questionnaire data. We then restrict the sample to adults (age ≥ 20) to define the analytic cohort used in subsequent analyses. A small proportion of records have missing values in BMI and diabetes status, which will be addressed later through multiple imputation.

Preview of merged NHANES 2013–2014 dataset limited to analysis variables (source columns only).

RIDAGEYR	BMXBMI	RIAGENDR	RIDRETH1	DIQ010
69	26.7	1	4	1
54	28.6	1	3	1
72	28.9	1	3	1
9	17.1	1	3	2

Preview of merged NHANES 2013–2014 dataset limited to analysis variables (source columns only).

RIDAGEYR	BMXBMI	RIAGENDR	RIDRETH1	DIQ010
73	19.7	2	3	2
56	41.7	1	1	2
0	NA	1	3	NA
61	35.7	2	3	2
42	NA	1	2	2
56	26.5	2	3	2

Variable Definitions

- **Response Variable:**
diabetes_dx (binary) represents a Type 2 diabetes diagnosis, excluding gestational diabetes. It was derived from DIQ010 (“Doctor told you have diabetes”), while DIQ050 (insulin use) was excluded to prevent treatment-related confounding.
- **Predictor Variables:**
 - BMXBMI – Body Mass Index (kg/m^2), treated as continuous and categorized into six BMI classes (bmi_cat).
 - RIDAGEYR – Age (continuous, 20–80 years)
 - RIAGENDR – Sex (factor, two levels)
 - RIDRETH1 – Ethnicity (factor, five levels)

Study Design and Survey-Weighted Analysis

The National Health and Nutrition Examination Survey (NHANES) is a nationally representative health survey conducted by the National Center for Health Statistics (NCHS), Centers for Disease Control and Prevention (CDC). NHANES utilizes a complex, multistage probability sampling design that incorporates stratification, clustering, and oversampling of specific population subgroups, including racial and ethnic minorities and older adults, to provide reliable estimates of health conditions among the noninstitutionalized civilian population of the United States (?).

Because NHANES data are collected using a complex survey design rather than simple random sampling, appropriate survey design variables were incorporated into the analysis. Primary sampling units (SDMVPSU), stratification variables (SDMVSTRA), and examination sample weights (WTMEC2YR) were retained to account for unequal probabilities of selection, nonresponse, and oversampling. Incorporating these design elements allows for unbiased population estimates, appropriate variance estimation, and valid statistical inference representative of U.S. adults (?).

Table 1: Variable descriptions for the adult analytic NHANES 2013–2014 dataset

Variable	Description	Type	Source
Analysis Variables			
diabetes_dx	Type 2 diabetes diagnosis (1 = Yes, 0 = No), derived from DIQ010; gestational diabetes excluded.	Categorical	NHANES DIQ010
age	Age in years.	Continuous	NHANES RIDAGEYR
bmi	Body mass index (kg/m ²) calculated from measured height and weight.	Continuous	NHANES BMXBMI
bmi_cat	BMI categories: Underweight, Normal weight, Overweight, and Obesity Classes I–III. Normal weight was used as the reference category.	Categorical	Derived from BMI
sex	Participant sex (Male, Female).	Categorical	NHANES RIAGENDR
race	Race/ethnicity categorized as White, Black, Hispanic, and Other.	Categorical	Derived from RIDRETH1
Survey Design Variables			
WTMEC2YR	Two-year examination sample weight for Mobile Examination Center participants.	Weight	NHANES design
SDMVPSU	Primary sampling unit identifier used for variance estimation.	Design	NHANES design
SDMVSTRA	Stratum identifier used for complex survey variance estimation.	Design	NHANES design
Derived Variables			
age_c	Centered and standardized age (z-score).	Continuous	Derived from age
bmi_c	Centered and standardized BMI (z-score).	Continuous	Derived from BMI

A survey-weighted logistic regression model was initially specified as a design-based analytical approach to evaluate the association between diabetes status and selected demographic and clinical predictors. The binary outcome variable, diabetes mellitus diagnosis (**diabetes_dx**), was derived from the NHANES Diabetes Questionnaire variable DIQ010 (“Doctor told you have diabetes”) and coded as 1 for participants with a diabetes diagnosis and 0 for participants without diabetes. Participants with gestational diabetes were excluded, and DIQ050 (insulin use) was not included in outcome classification to minimize potential treatment-related confounding.

The primary covariates included established diabetes risk factors:

- **Age:** Participant age in years (**age**), analyzed as a continuous variable and standardized as **age_c** for modeling.
- **Body Mass Index (BMI):** Measured BMI (**bmi**) calculated from examination-based height and weight measurements. BMI was evaluated as a continuous standardized variable (**bmi_c**) and categorized into clinically relevant BMI groups (**bmi_cat**).
- **Sex:** Biological sex obtained from NHANES demographic data (**sex**) and categorized as male or female.
- **Race/Ethnicity:** Race and ethnicity (**race**) were included as categorical predictors and collapsed into four analytical categories: White, Black, Hispanic, and Other.

The survey-weighted analytical framework accounted for the NHANES complex sampling structure and provided population-representative estimates of diabetes associations among U.S. adults. The results from this design-based model were used as a comparative framework for evaluating the performance and interpretability of the Bayesian logistic regression approach.

Step	Description
Weighting	Used the survey package to calculate weighted means for key variables (e.g., age and diabetes status) and to estimate design effects and effective sample size for the complex survey design.
Standardization	Centered and standardized BMI and age (bmi_c , age_c) for use in regression models.
Age	Not implemented in the analytic dataset (continuous age retained).
Categorization	Reference retained for potential descriptive grouping (20–<30, 30–<40, 40–<50, 50–<60, 60–<70, 70–80).
BMI	Recoded as: <18.5 (Underweight), 18.5–<25 (Normal), 25–<30 (Overweight), 30–<35 (Obesity I), 35–<40 (Obesity II), 40 (Obesity III).

Step	Description
Ethnicity Recoding	RIDRETH1 recoded as: 1 = Mexican American, 2 = Other Hispanic, 3 = Non-Hispanic White, 4 = Non-Hispanic Black, 5 = Other/Multi; then NH White set as the reference level (five analytical levels retained).
Special Codes	Transformed nonresponse codes (e.g., 3, 7, 9) to NA. These missing codes were evaluated for potential nonrandom patterns (MAR/MNAR).
Missing Data	Retained and visualized missing values (primarily in BMI and diabetes status) to assess their pattern and informativeness before multiple imputation.
Final Dataset	Created the cleaned analytic dataset (adult) using <i>Non-Hispanic White</i> and <i>Male</i> as reference groups for modeling, preserving NHANES survey design variables (WTMEC2YR, SDMVPSU, SDMVSTRA).

Adult Cohort (NHANES 2013–2014)

This analysis uses data from the 2013–2014 cycle of the National Health and Nutrition Examination Survey (NHANES), a nationally representative survey of U.S. adults. Individuals aged **20 years and older** with available diabetes outcome data and key demographic and anthropometric variables were included in the analytic cohort.

Three NHANES components were merged to construct the dataset:

- **DEMO_H**: demographic characteristics and sample design variables
- **BMX_H**: measured height and weight used to calculate BMI
- **DIQ_H**: doctor-diagnosed diabetes status

The combined dataset contained **n = 5,769 adults** after applying inclusion criteria and preparing variables for analysis. The primary outcome was **doctor-diagnosed diabetes** (diabetes_dx). Predictor variables included:

- **Age**, standardized as a z-score for interpretability
- **BMI**, standardized as a z-score
- **Sex** (Female vs. Male)
- **Race/Ethnicity**, with Non-Hispanic White as the reference group

Survey design features—strata, primary sampling units, and sample weights—were retained to support design-based logistic regression and ensure appropriate population-level inference.

Adult Cohort Definition

Adult Cohort Definition and Survey-Weighted Characteristics

The analytic cohort consisted of adult NHANES 2013–2014 participants aged ≥ 20 years with available information on diabetes status and selected predictor variables. Survey-weighted analyses incorporated the NHANES complex sampling design using examination sample weights (WTMEC2YR), primary sampling units (SDMVPSU), and strata (SDMVSTRA). This approach accounted for unequal probabilities of selection, nonresponse, and oversampling to generate nationally representative estimates for the U.S. adult population.

The survey-weighted mean age of participants was 47.5 years (SE = 0.38). The weighted prevalence of diabetes was 8.9% (SE = 0.005), indicating that approximately 9% of the adult NHANES population met the diabetes classification criteria.

The complex survey design substantially influenced variance estimation. The design effect for the diabetes outcome was estimated as 4759.91, reflecting the impact of clustering, stratification, and weighting on variance relative to a simple random sample. The corresponding effective sample size was estimated as 48,142 individuals after accounting for the survey design effects.

Descriptive characteristics of demographic and clinical variables in the adult NHANES 2013–2014 analytic cohort are presented in Tables ?? and ?. Continuous variables, including age and body mass index (BMI), are summarized using mean, standard deviation (SD), range, and missingness. Categorical variables are presented as frequencies and percentages. summarized in Table ?.

Table 1a. Continuous variables (age, BMI): N, missing, mean (SD), range.

Variable	N	Missing	Mean	SD	Min	Max
age	5769	0	49.11	17.56	20.0	80.0
bmi	5520	249	29.10	7.15	14.1	82.9

Table 1b. Categorical variables (sex, race, diabetes_dx, bmi_cat): counts and percentages.

Variable	Level	n	pct
bmi_cat	25–<30	1768	30.6
bmi_cat	18.5–<25	1579	27.4
bmi_cat	30–<35	1145	19.8
bmi_cat	35–<40	519	9.0
bmi_cat	40	419	7.3
bmi_cat	(Missing)	249	4.3
bmi_cat	<18.5	90	1.6
diabetes_dx	No	4974	86.2
diabetes_dx	Yes	618	10.7

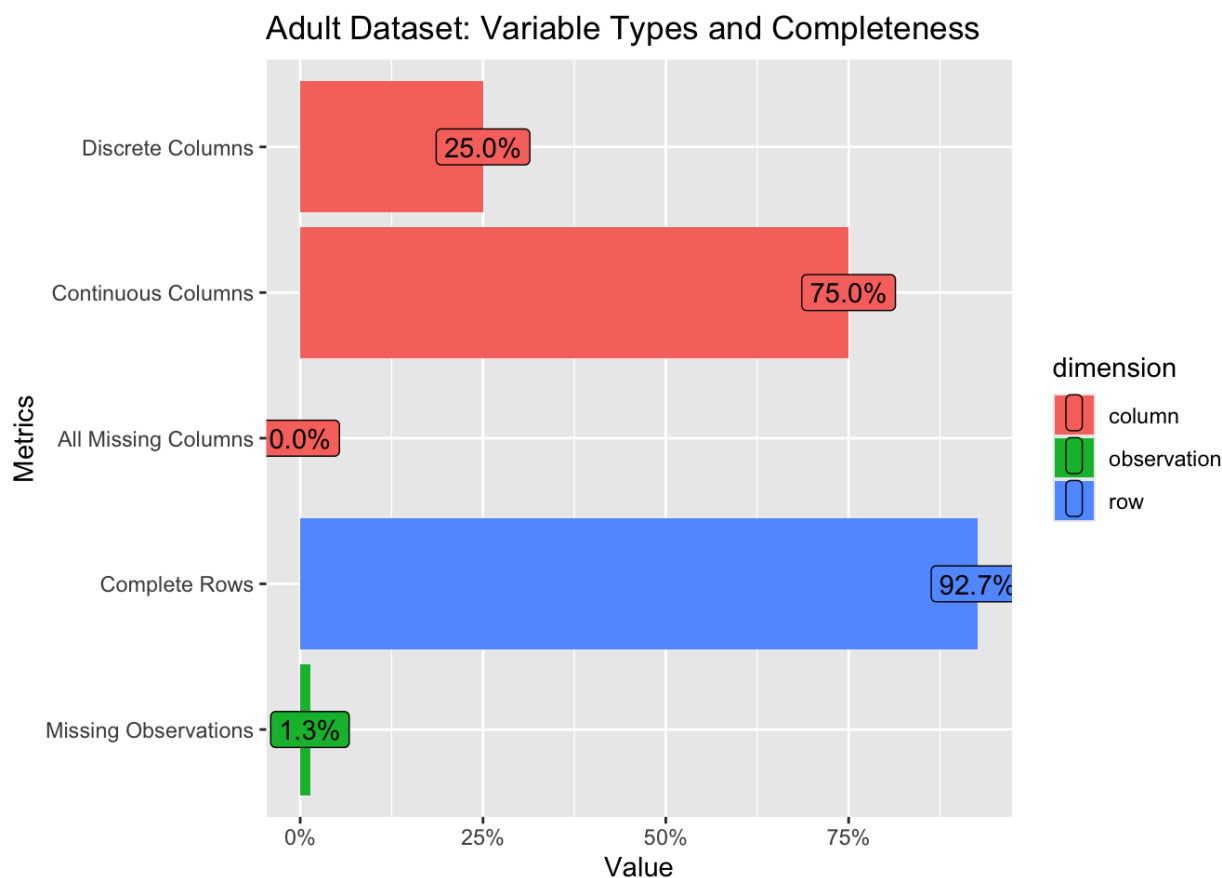
Table 1b. Categorical variables (sex, race, diabetes_dx, bmi_cat): counts and percentages.

Variable	Level	n	pct
diabetes_dx	(Missing)	177	3.1
race	NH White	2472	42.8
race	NH Black	1177	20.4
race	Other/Multi	845	14.6
race	Mexican American	767	13.3
race	Other Hispanic	508	8.8
sex	Female	3011	52.2
sex	Male	2758	47.8

Table 1a and 1b summarize the analytic variables included in subsequent models. Mean age and BMI values indicate an adult cohort spanning a wide range of body composition, while categorical summaries confirm balanced sex representation and sufficient sample sizes across race/ethnicity categories. These variables were standardized and used as predictors in all modeling frameworks (analytic cohort N = 5,769 adults 20 years).

The analytic adult cohort (N = 5,769) includes standardized variables for age and BMI (age_c, bmi_c), categorical indicators for sex and race/ethnicity (race), and a binary doctor-diagnosed diabetes variable (diabetes_dx). This is a clean dataset with all NAs removed.

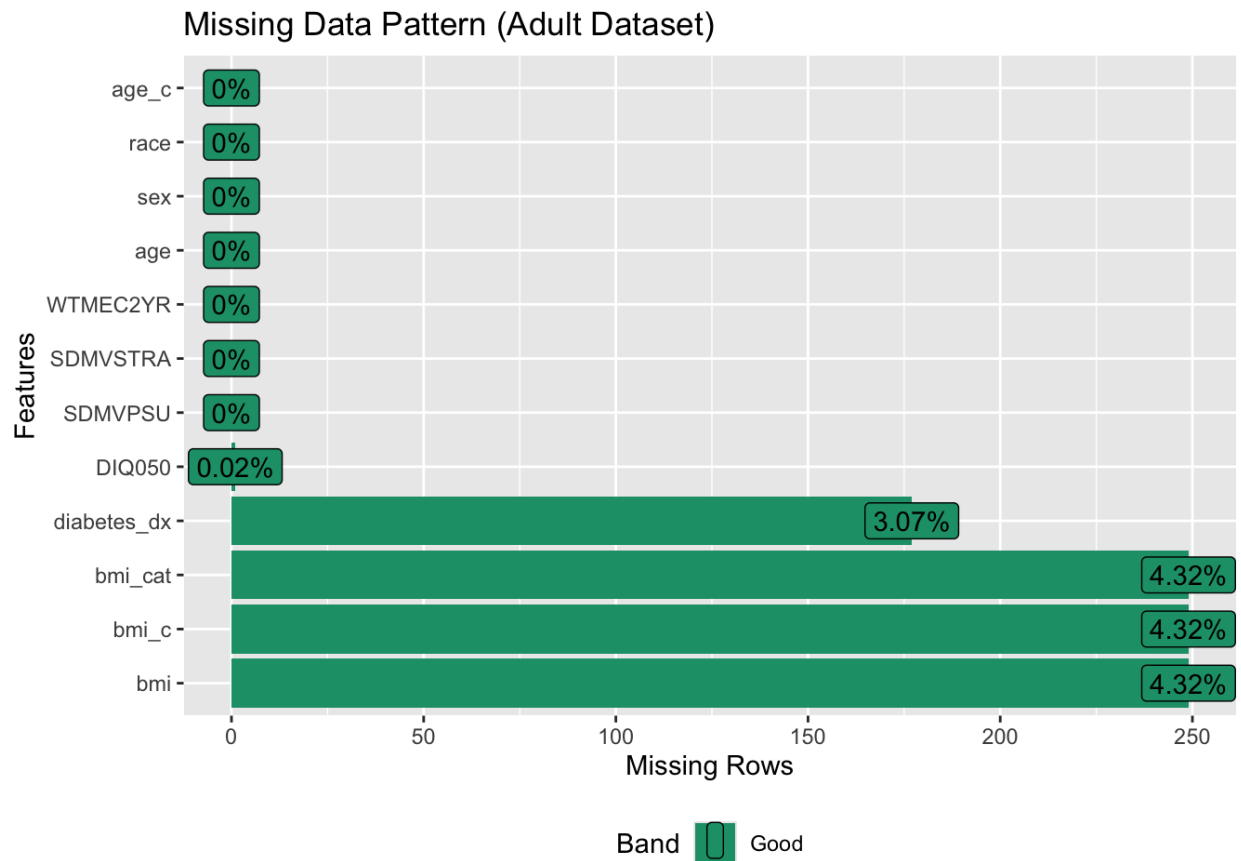
plot_intro(adult, title = "Adult Dataset: Variable Types and Completeness")



The visual overview indicates that 75% of variables are continuous and 25% are categorical, with no completely missing columns. Approximately 92.7% of rows are fully complete, and only 1.3% of observations contain missing values, suggesting minimal data loss prior to imputation.

Missing Data Summary

Missing data were minimal across analytic variables. BMI-related fields (bmi, bmi_c, bmi_cat) showed ~4.3% missingness, and diabetes_dx showed ~3.1%. All demographic and survey design variables were complete, indicating that missingness was limited to health-related measures and appropriate for multiple imputation. Overall missingness was low (~7.3%). Gaps were concentrated in bmi (n = 249) and diabetes_dx (n = 177), while demographic and design variables were complete. This limited pattern of missingness is consistent with a Missing At Random (MAR) mechanism and likely reflects reduced participation in the physical examination component among certain adults.



Missingness for Key Analysis Variables.

Variable	Missing_n	Missing_pct
bmi	249	4.3
diabetes_dx	177	3.1

Exploratory Data Analysis (EDA)

Missing Data

Overall missingness in the analytic variables was low, with approximately **3–4%** missing data in age or BMI. This level of missingness supported the use of multiple imputation during data preparation and ensured that complete-case analyses remained consistent for model comparison.

Outcome Distribution

Doctor-diagnosed diabetes prevalence in the adult cohort was approximately **11%**, which aligns with national estimates for U.S. adults during the same time period.

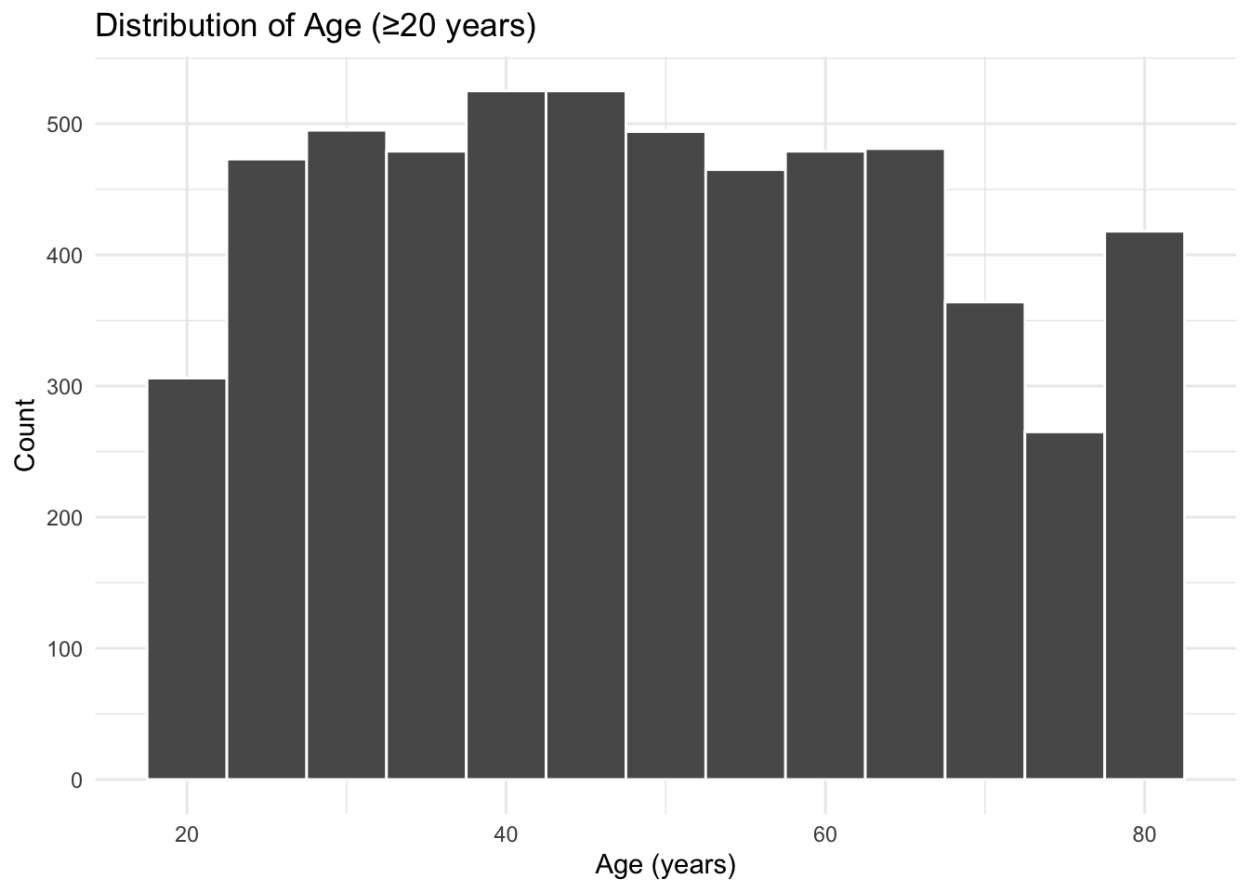
Predictor Patterns

Exploratory analyses revealed several expected patterns:

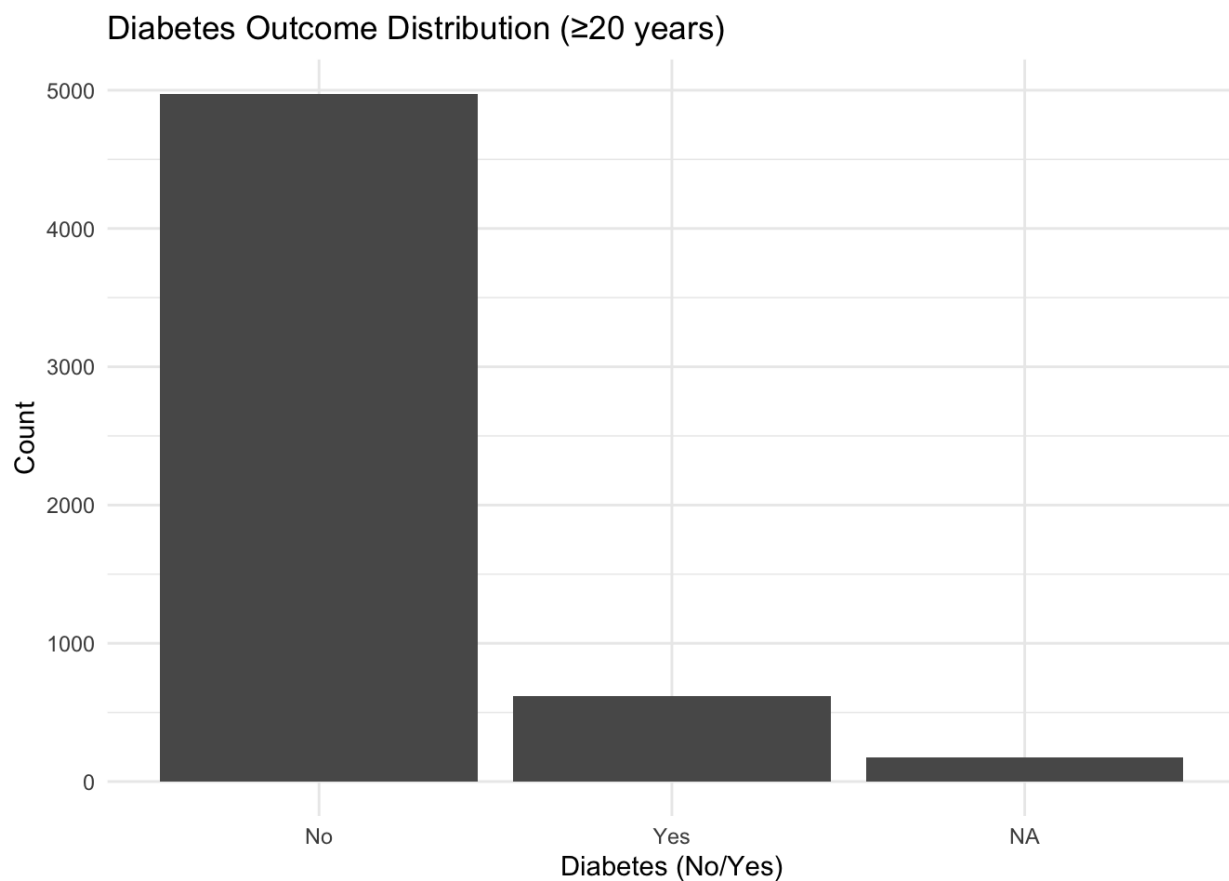
- **Age:** Diabetes prevalence increased steadily across older age groups.
- **BMI:** Individuals with diabetes tended to have higher BMI values, with clear separation between diabetes and non-diabetes groups.
- **Sex:** Females exhibited slightly lower diabetes prevalence compared to males.
- **Race/Ethnicity:** Prevalence was higher among Non-Hispanic Black, Hispanic, and “Other” racial/ethnic groups compared to Non-Hispanic White adults.

These distributional patterns reflect established epidemiologic relationships and support the suitability of the dataset for logistic regression modeling.

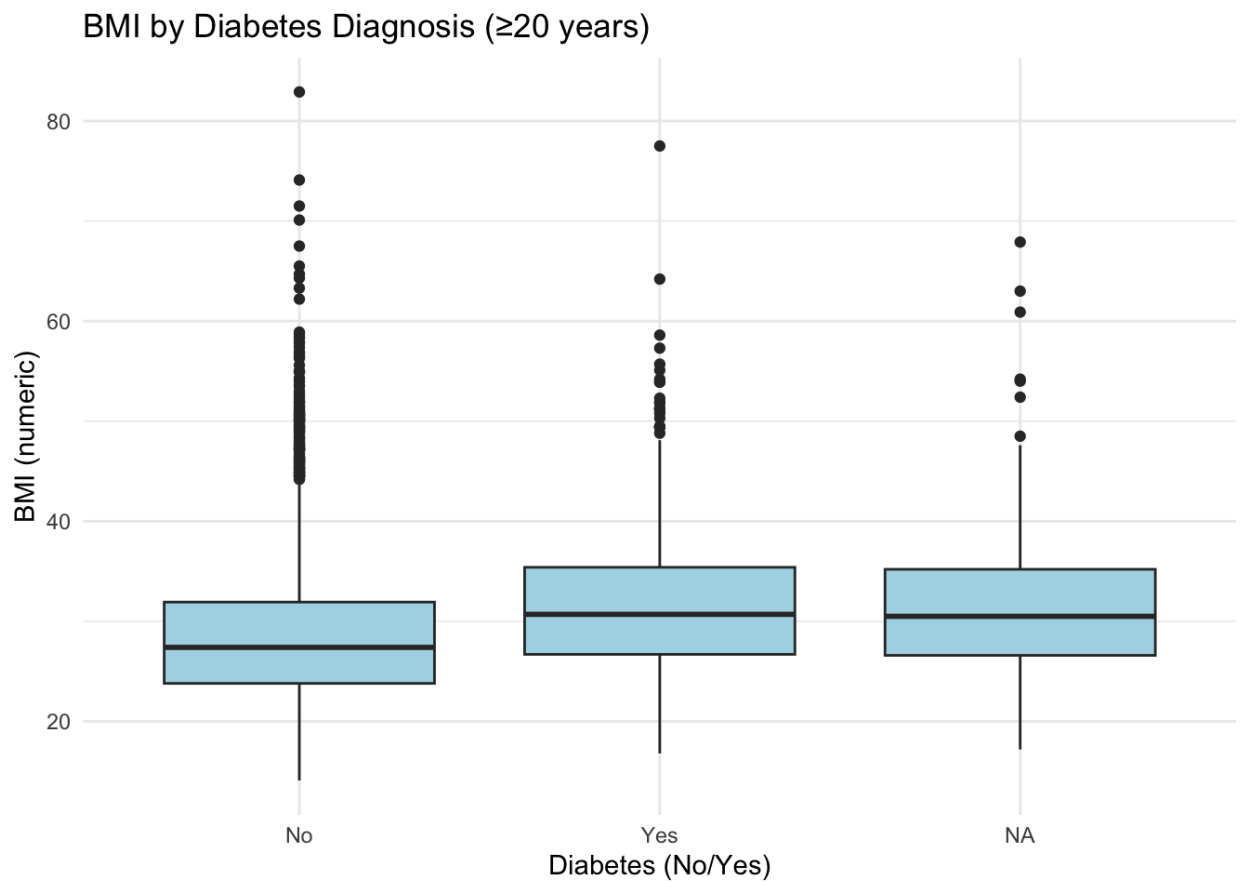
EDA Visuals



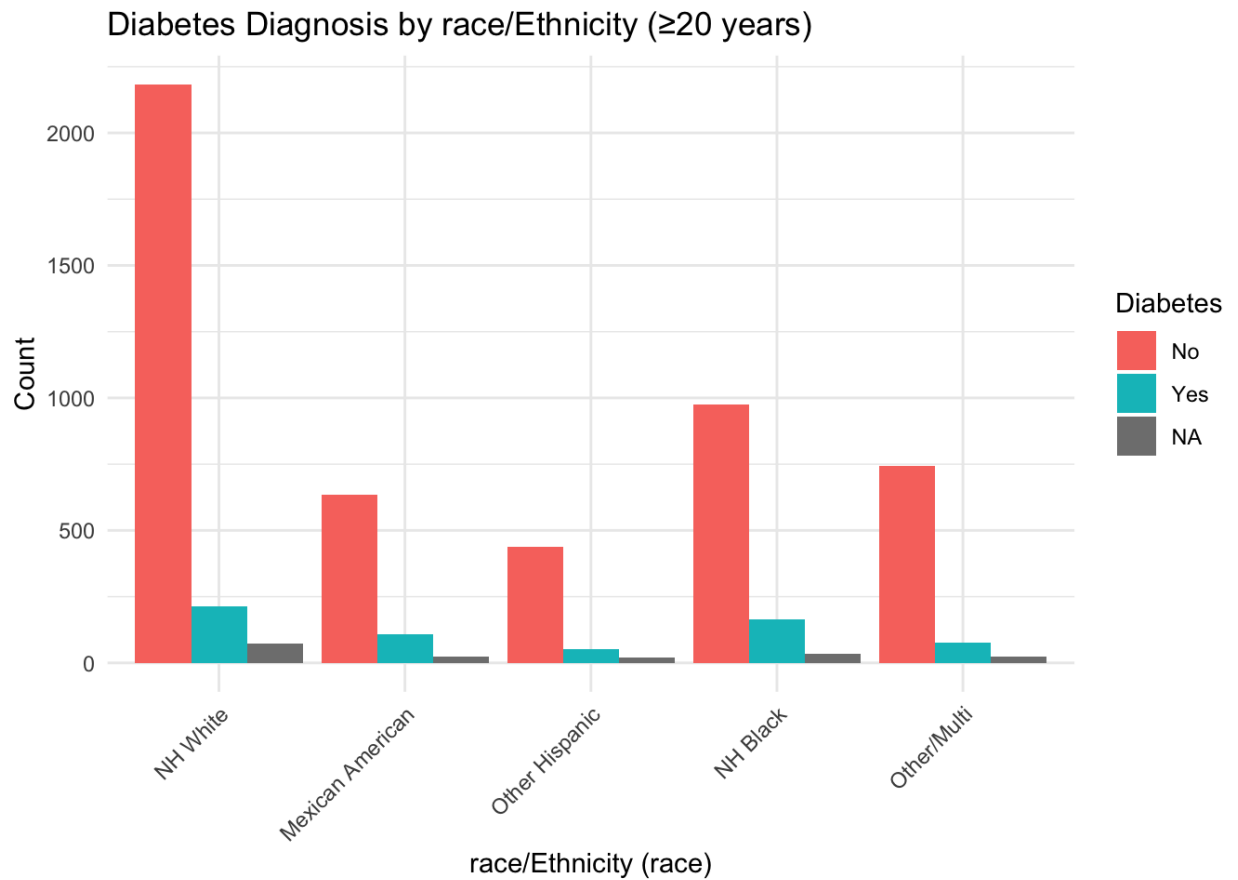
Distribution of age among adults aged 20 years. The sample spans 20–80 years, with peak representation between 30 and 50 years and a gradual decline in older age groups, reflecting a balanced adult cohort suitable for regression modeling.



Distribution of diabetes outcomes among adults aged 20 years. Most participants reported no diabetes diagnosis (No), while approximately 11% had diabetes (Yes) and 3% had missing responses, reflecting expected population prevalence and limited outcome missingness.



Distribution of BMI by diabetes diagnosis among adults aged ≥ 20 years. Participants with diabetes (Yes) show a higher median BMI and greater variability compared to those without diabetes (No), supporting the established positive association between obesity and diabetes risk.



Diabetes diagnosis by race/ethnicity among adults aged ≥ 20 years. Non-Hispanic Black and Hispanic participants show higher proportions of diabetes diagnoses compared with Non-Hispanic White participants, reflecting known disparities in diabetes prevalence across racial and ethnic groups.

Modeling Frameworks

Two modeling frameworks were compared using identical predictors—standardized age, BMI, sex, and race—and the binary outcome `diabetes_dx`:

1. Survey-weighted logistic regression to account for the NHANES complex sampling design,
2. Bayesian logistic regression with weakly informative priors to quantify parameter uncertainty.

Survey-Weighted Logistic Regression

The NHANES 2013–2014 data use a complex, multistage probability design involving strata (`SDMVSTRA`), primary sampling units (PSUs; `SDMVPSU`),

Table 2: Structure of the final analytic dataset used for Bayesian logistic regression

Characteristic	Value	Description
Number of observations	5,349	Adult participants with complete data
Number of variables	5	Variables included in the final model
diabetes_dx	Numeric (0/1)	Diabetes status (0 = No, 1 = Yes)
sex	Factor (2 levels)	Male, Female
race	Factor (5 levels)	NH White, Mexican American, Other Hispanic, NH Black, Other
age_c	Numeric	Centered age (continuous)
bmi_c	Numeric	Centered body mass index (continuous)

and examination weights (WTMEC2YR) to ensure nationally representative estimates (National Center for Health Statistics (NCHS) 2014).

Estimates are population-weighted using NHANES survey design variables (WTMEC2YR, SDMVSTRA, SDMVPSU). Odds ratios are reported per one standard deviation (1 SD) increase in age and BMI, with reference groups Male and White.

Below is a structure of the analytic dataset used for regression modeling, showing variable names, types, and sample values (N = 5,349).

The cleaned analytic dataset used for Bayesian logistic regression consisted of 5,349 participants after excluding observations with missing values required for model estimation. The outcome variable, **diabetes_dx**, was coded as a binary numeric variable (0 = no diabetes, 1 = diabetes). Predictor variables included sex (2 levels), race/ethnicity (5 levels), and standardized continuous variables for age (**age_c**) and body mass index (**bmi_c**). The structure of the analytic dataset is summarized in Table ??.

```
svyglm(formula = diabetes_dx ~ age_c + bmi_c + sex + race, design =
nhanes_design_adult,
```

```
family = quasibinomial())
```

Survey design:

```
survey::svydesign(id = ~SDMVPSU, strata = ~SDMVSTRA, weights = ~WT-
MEC2YR, nest = TRUE, data = adult_clean)
```

Table 1: Survey-weighted logistic regression: odds ratios (OR) and 95% confidence intervals for diabetes diagnosis among adults (NHANES 2013–2014).

term	OR	LCL	UCL	p.value
age_c	3.0292807	2.6967690	3.4027912	0.0000000
bmi_c	1.8853571	1.6526296	2.1508579	0.0000039
sexFemale	0.5281132	0.4104905	0.6794397	0.0003857

term	OR	LCL	UCL	p.value
raceMexican American	2.0358434	1.4850041	2.7910081	0.0008262
raceOther Hispanic	1.5915182	1.1664529	2.1714810	0.0087119
raceNH Black	1.6689718	1.1605895	2.4000450	0.0116773
raceOther/Multi	2.3270527	1.5451752	3.5045697	0.0014331

Interpretation

age_c and **bmi_c** emerge as the strongest predictors of diabetes in the NHANES 2013–2014 adult cohort.

- For **age_c**, each 1-SD increase is associated with an odds ratio of **3.03** (95% CI: 2.70–3.40), indicating that older adults have **nearly three times higher odds** of diagnosed diabetes compared to younger adults.
- For **bmi_c**, each 1-SD increase corresponds to an odds ratio of **1.89** (95% CI: 1.65–2.15), meaning higher BMI substantially elevates diabetes risk.

Sex differences were also observed.

- Compared to males, females had significantly lower odds of diabetes (OR = **0.53**, 95% CI: 0.41–0.68), consistent with known sex-related differences in metabolic and cardiometabolic profiles.

Racial and ethnic disparities in diabetes risk were pronounced.

Using Non-Hispanic Whites as the reference group:

- **Mexican American** adults had about **2.04×** higher odds of diabetes (95% CI: 1.49–2.79).
- **Other Hispanic** adults showed **1.59×** higher odds (95% CI: 1.17–2.17).
- **Non-Hispanic Black** adults had **1.67×** higher odds (95% CI: 1.16–2.40).
- **Other/Multi-racial** adults exhibited **2.33×** higher odds (95% CI: 1.55–3.50).

All predictors were statistically significant ($p < 0.05$), and the direction and magnitude of effects reflect well-established demographic and clinical patterns in diabetes risk.

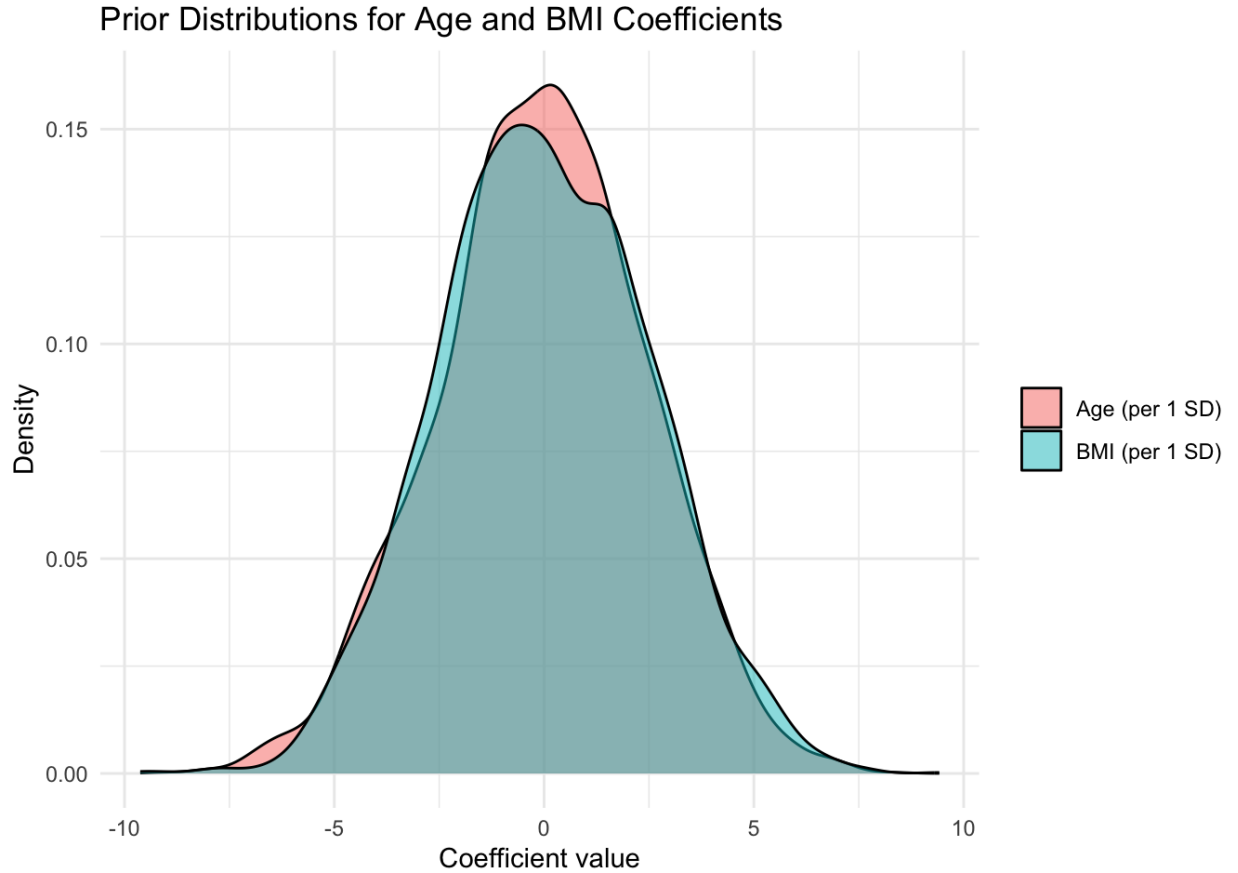
Bayesian Logistic Regression

Bayesian logistic regression was used to quantify parameter uncertainty and compare posterior estimates with the survey-weighted model. Weakly informative priors were applied to regularize estimates while preserving flexibility in inference.

Model Specifications: - **Family:** Bernoulli with logit link

- **Data:** adult_imp1 (N = 5,592)
- **Chains:** 4 (2,000 iterations each; 1,000 warmup)
- **Adaptation delta:** 0.95
- **Weights:** Normalized NHANES examination weights (wt_norm, mean = 1.00, SD = 0.79)
- **Predictors:** Standardized age, BMI, sex, and race

Define Model and Priors



Prior distributions for standardized age and BMI coefficients, assuming $\text{Normal}(0, 2.5)$ priors. These weakly informative priors constrain extreme coefficient values while allowing flexibility in posterior estimation, ensuring regularization without strong bias.

Data Preparation Using Multiple Imputation by Chained Equations (MICE)

Multiple Imputation by Chained Equations (MICE) was used to address missing data in the analytic dataset (Stef van Buuren and Groothuis-Oudshoorn 2011; S. van Buuren 2012). MICE iteratively imputes each incomplete variable using regression models based on the available data, producing multiple completed datasets that incorporate uncertainty from the imputation process. Estimates across imputations are then combined using Rubin's rules to obtain final parameter estimates and confidence intervals.

As an alternative to full Bayesian joint modeling, MICE offers a flexible and robust approach for handling missingness through chained regression equations. For large sample sizes ($n = 400$), even when a single variable contains substantial

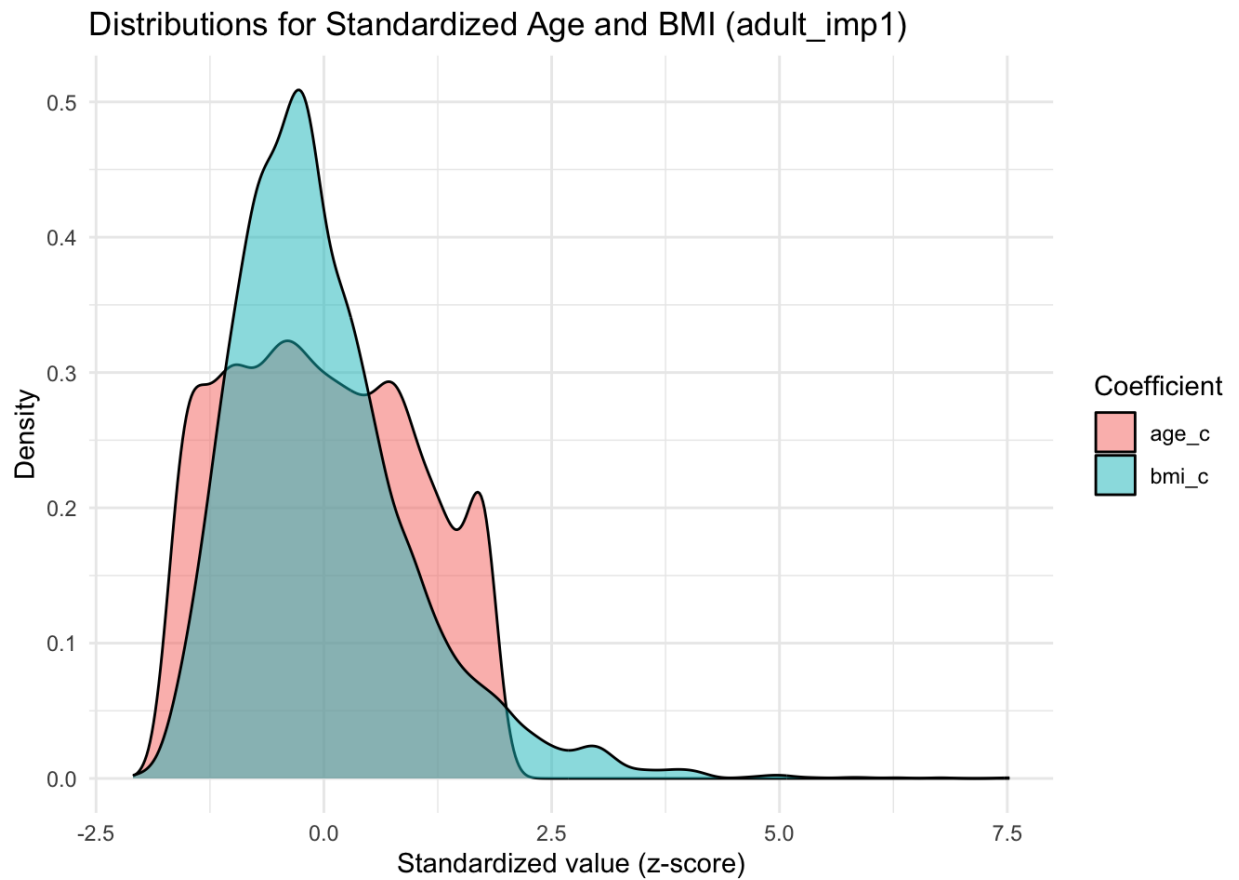
missingness (up to 75%), MICE performs reliably under skewed, multimodal, or heavy-tailed distributions without materially affecting mean-structure estimation (S. van Buuren 2012).

In this study, continuous variables were imputed using regression-based methods (age via normal linear regression and BMI via predictive mean matching to preserve distributional shape). Categorical variables (sex and race) were imputed using logistic and polytomous regression models. The outcome variable (diabetes_dx) was **not** imputed and was used only as a predictor during the imputation process. Twenty imputations were generated to reduce Monte Carlo error and ensure stable variance estimation.

The chained-equations process demonstrated stable convergence across iterations, supporting reliable estimation of missing BMI values (and age, where applicable). After imputation, the completed analytic dataset contained **n = 5,592 adults** with all key predictors available for downstream modeling, including the Bayesian logistic regression framework.

'data.frame': 5592 obs. of 11 variables:

```
$ diabetes_dx: num 1 1 1 0 0 0 0 0 0 0 ...
$ age : num 69 54 72 73 56 61 42 56 65 26 ...
$ bmi : num 26.7 28.6 28.9 19.7 41.7 35.7 23.6 26.5 22 20.3 ...
$ sex : Factor w/ 2 levels "Male","Female": 1 1 1 2 1 2 1 2 1 2 ...
$ race : Factor w/ 5 levels "NH White","Mexican American",...: 4 1 1 1 2 1 3 1 1 1 ...
$ WTMEC2YR : num 13481 24472 57193 65542 25345 ...
$ SDMVPSU : num 1 1 1 2 1 1 2 1 2 2 ...
$ SDMVSTRA : num 112 108 109 116 111 114 106 112 112 113 ...
$ age_c : num 1.132 0.278 1.303 1.36 0.392 ...
$ bmi_c : num -0.3332 -0.0676 -0.0256 -1.3118 1.7639 ...
$ wt_norm : num 0.339 0.616 1.44 1.65 0.638 ...
```



Distribution of standardized age (age_c) and BMI (bmi_c) in the imputed dataset (adult_imp1). Both variables were mean-centered and scaled (z-scores) for inclusion in regression models. The overlapping density curves indicate approximate normality and comparable variance, supporting suitability for standardized coefficient estimation.

After applying exclusions and resolving the small amount of missingness (3–4%), both the survey-weighted and Bayesian models were fit to the same complete-case adult dataset (n = 5,769).

Fit the Model

Running MCMC with 4 sequential chains...

Chain 1 finished in 11.2 seconds.

Chain 2 finished in 10.4 seconds.

Chain 3 finished in 10.8 seconds.

Chain 4 finished in 11.5 seconds.

All 4 chains finished successfully.

Mean chain execution time: 11.0 seconds.

Total execution time: 44.3 seconds.

Family: bernoulli

Links: mu = logit

Formula: diabetes_dx | weights(wt_norm) ~ age_c + bmi_c + sex + race

Data: adult_imp1 (Number of observations: 5592)

Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;

total post-warmup draws = 4000

— coefficient table —

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Bayesian logistic regression model fit summary for diabetes diagnosis (diabetes_dx) with standardized predictors (age, BMI, sex, and race) and normalized NHANES weights. All four MCMC chains (4,000 post-warmup draws) converged successfully (R = 1.00), indicating stable estimation across parameters.

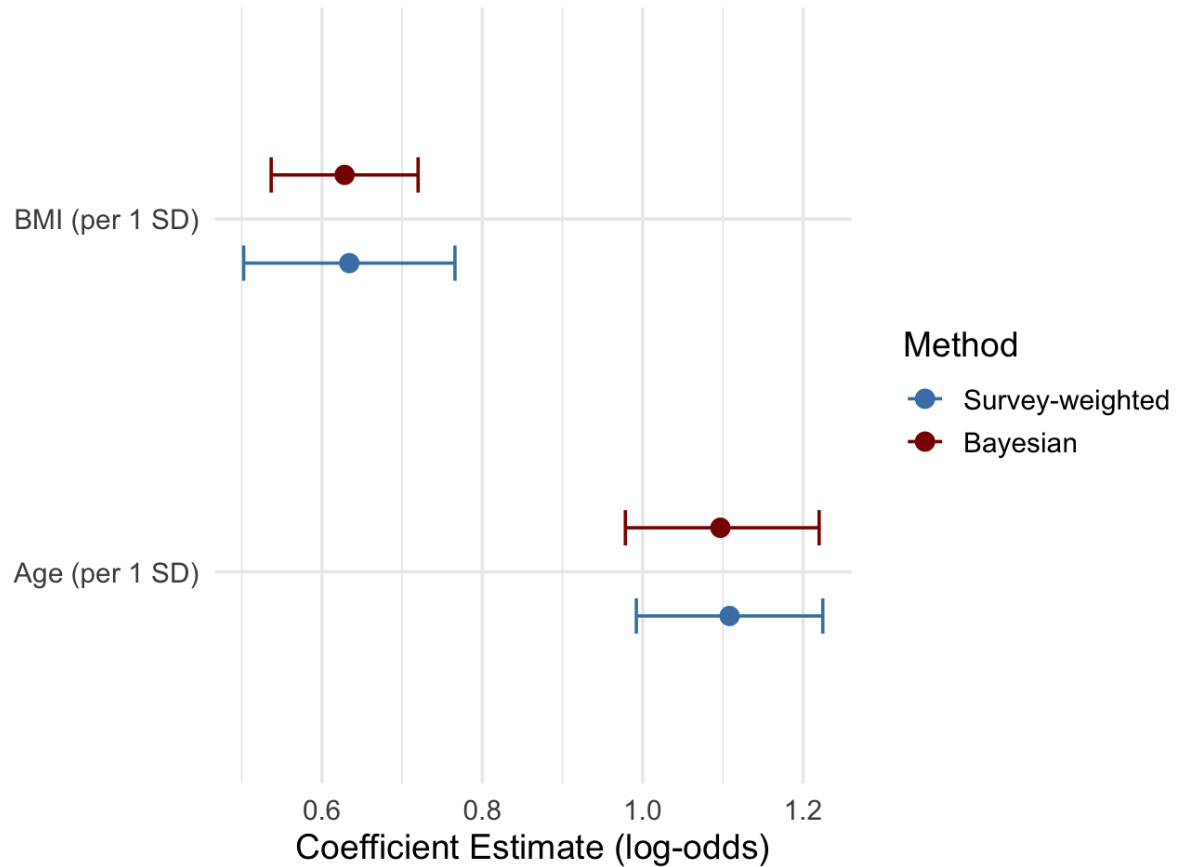
Table 3: Odds Ratios (95% CI) for Key Predictors Across Survey-weighted and Bayesian Models.

Odds Ratios (95% CI) Across Survey-weighted and Bayesian Models		
term	Survey-weighted	Bayesian
Age (per 1 SD)	3.03 (2.70 – 3.40)	2.99 (2.66 – 3.39)
BMI (per 1 SD)	1.89 (1.65 – 2.15)	1.87 (1.71 – 2.05)
Black (vs. White)	1.67 (1.16 – 2.40)	1.70 (1.26 – 2.30)
Female (vs. Male)	0.53 (0.41 – 0.68)	0.52 (0.42 – 0.63)
Hispanic (vs. White)	1.59 (1.17 – 2.17)	1.53 (0.94 – 2.43)
Mexican American (vs. White)	2.04 (1.49 – 2.79)	1.99 (1.41 – 2.80)
Other/Multi (vs. White)	2.33 (1.55 – 3.50)	2.26 (1.56 – 3.24)

- Age and BMI show strong positive associations with diabetes (credible intervals exclude 1).
- Female sex shows lower odds than male (protective factor).
- Non-White racial groups have higher odds compared with Whites, consistent with known disparities.

- All model parameters exhibit well-defined, unimodal posteriors with narrow credible intervals.

Coefficient Comparison Plot



Coefficient estimates (log-odds) from survey-weighted and Bayesian logistic regression models for Age and BMI.

The coefficient comparison plot shows that both modeling approaches identify Age and BMI as strong positive predictors of diabetes. The survey-weighted and Bayesian estimates are closely aligned, with overlapping intervals, indicating consistent effect sizes across methods. This agreement supports the robustness of the findings regardless of whether weighting or Bayesian regularization is used.

Posterior Coefficients (log-odds scale)

Intercept

- **b_Intercept** = **-2.66** (SD = 0.09, 95% CrI: -2.83 to -2.50)
- Represents baseline log-odds of diabetes for:

Male, Non-Hispanic White, average age, average BMI

- Corresponds to a baseline probability of ~**6.5%**.

Age (per 1 SD)

- **b_age_c** = **1.10** (SD = 0.06, 95% CrI: 0.98 to 1.22)
- Strong positive effect on diabetes risk.
- Odds ratio: **exp(1.10)** **2.99**
- A 1-SD increase in age **nearly triples the odds** of diabetes.
- Credible interval excludes 0 → strong evidence of association.

BMI (per 1 SD)

- **b_bmi_c** = **0.63** (SD = 0.05, 95% CrI: 0.54 to 0.72)
- Positive association with diabetes.
- Odds ratio: **exp(0.63)** **1.87**
- A 1-SD increase in BMI increases odds of diabetes by ~**87%**.
- Tight CrI → high certainty in the estimate.

Sex (Female vs. Male)

- **b_sexFemale** = **-0.66** (SD = 0.10, 95% CrI: -0.86 to -0.47)
- Indicates lower log-odds for females.
- Odds ratio: **exp(-0.66)** **0.52**
- Females have ~**48% lower odds** of diabetes compared to males, adjusting for age, BMI, and race.

Race/Ethnicity

(Reference: **Non-Hispanic White**)

Mexican American

- **b** = **0.69** (SD = 0.17, 95% CrI: 0.34 to 1.03)
- OR **1.99**
- Nearly twice the odds of diabetes.

Other Hispanic

- **b** = **0.43** (SD = 0.25, 95% CrI: -0.07 to 0.89)
- OR **1.54**
- Mean effect positive, but CrI crosses 0 → weaker evidence.

Non-Hispanic Black

- **b** = **0.53** (SD = 0.15, 95% CrI: 0.23 to 0.83)
- OR **1.71**
- ~70% higher odds vs NH Whites, strong evidence.

Other / Multiracial

- **b** = **0.81** (SD = 0.19, 95% CrI: 0.45 to 1.18)
- OR **2.27**
- More than double the odds compared to NH Whites.

Interpretation

- **Age and BMI** show strong positive associations with diabetes risk in the posterior distribution.
- **Female sex** shows a clear protective association.
- **Racial/ethnic groups**, particularly Mexican American, NH Black, and Other/Multi, have elevated odds relative to NH Whites.
- Posterior estimates demonstrate consistent directionality with survey-weighted models, with Bayesian credible intervals providing more intuitive uncertainty quantification.

Diagnostics and Model Fit

Table 4: Bayesian R² Summary

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.1316278	0.0123417	0.107432	0.1565549

Table 5: MCMC Diagnostics (R-hat and Effective Sample Sizes) for Model Parameters

Parameter	Rhat	Bulk_ESS	Tail_ESS
b_Intercept	1	3548.0	3511.8
b_age_c	1	2349.3	2617.8
b_bmi_c	1	3327.1	2825.9
b_sexFemale	1	3668.1	3123.7
b_raceMexicanAmerican	1	3656.6	2821.2
b_raceOtherHispanic	1	4242.3	3013.5
b_raceNHBlack	1	3809.1	3012.2
b_raceOtherDMulti	1	3947.9	2809.1

All parameters achieved R = 1.00 and effective sample sizes >2,000, indicating excellent convergence. The Bayesian R² = 0.13, showing that age, BMI, sex, and race explain about 13% of diabetes variability.

Model Comparison

Table 3: Bayesian model comparison using approximate leave-one-out cross-validation (LOO-CV).

Model	elpd_diff	SE(elpd_diff)	p_loo	LOOIC
Base model (age + BMI + sex + race)	0.00	0.00	8.73	2836.52
Without race	-14.54	6.36	5.35	2865.60
Without sex	-19.90	8.21	7.25	2876.32

Models excluding race or sex had lower expected log predictive density (elpd), confirming that both variables contribute meaningfully to model fit.

Posterior Predictive Checks

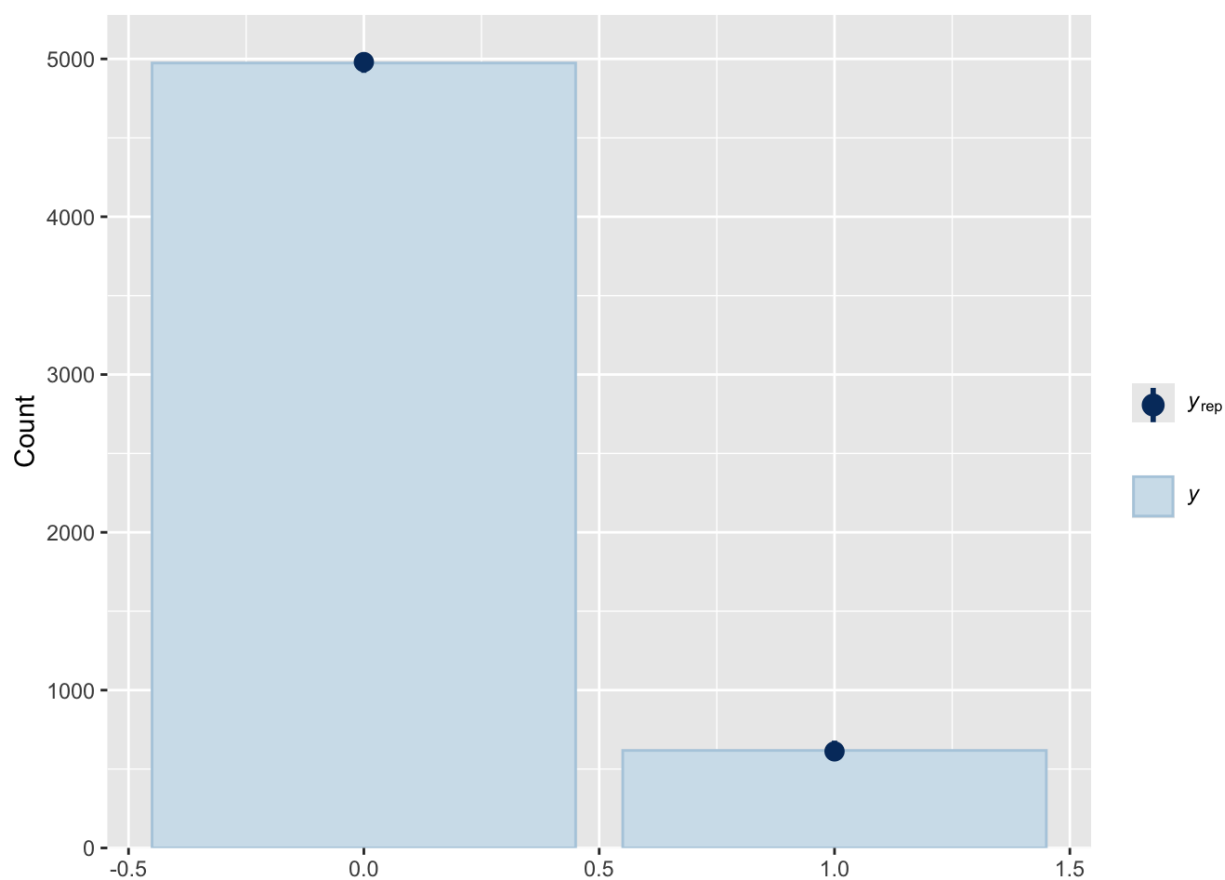


Figure 1: Posterior Predictive Check: Observed vs. Replicated Outcome Distribution (Bars)

The close alignment between observed (y) and replicated (y_rep) outcome

distributions indicates that the Bayesian model reproduces the empirical data structure well.

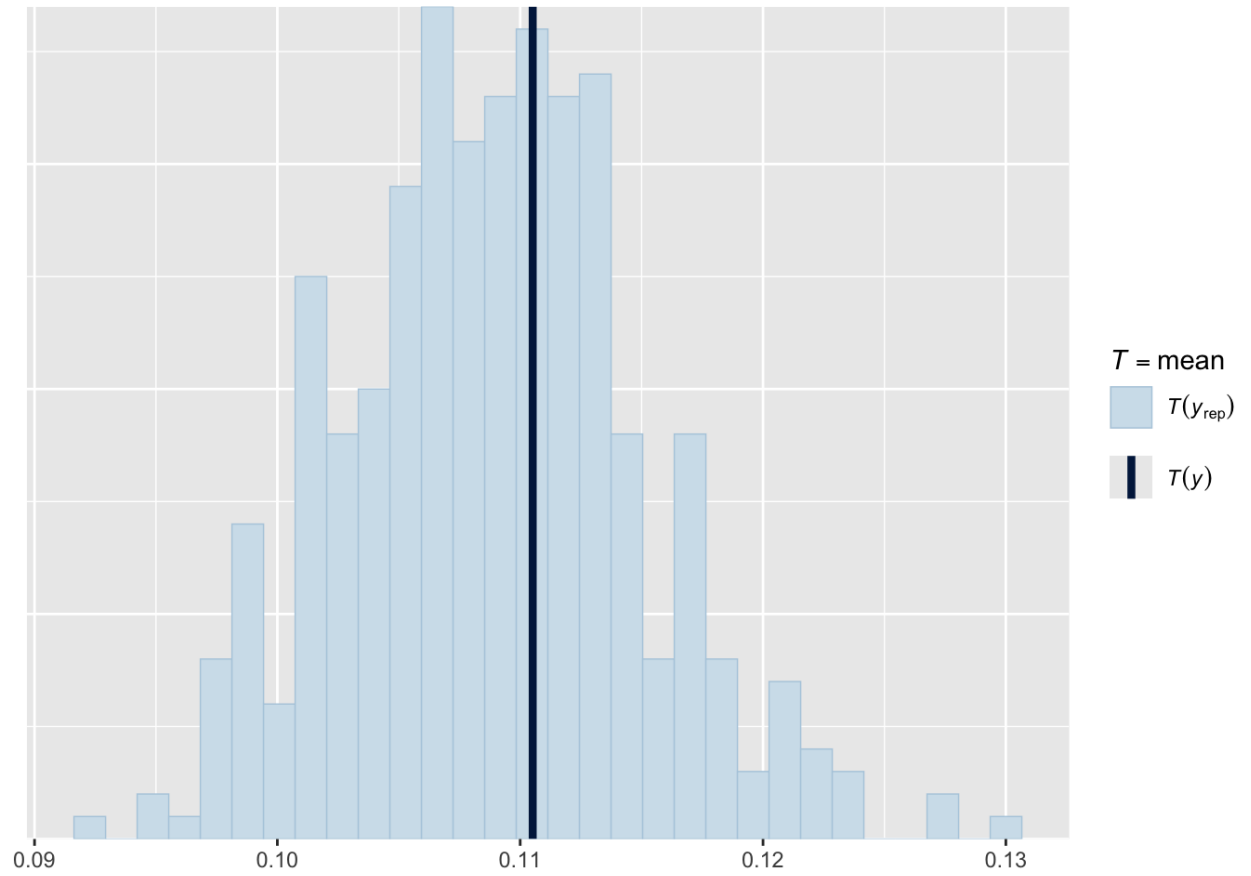


Figure 2: Posterior predictive check for the mean of the binary outcome, comparing the observed mean ($T(y)$) to replicated means ($T(y_{\text{rep}})$) across posterior draws.

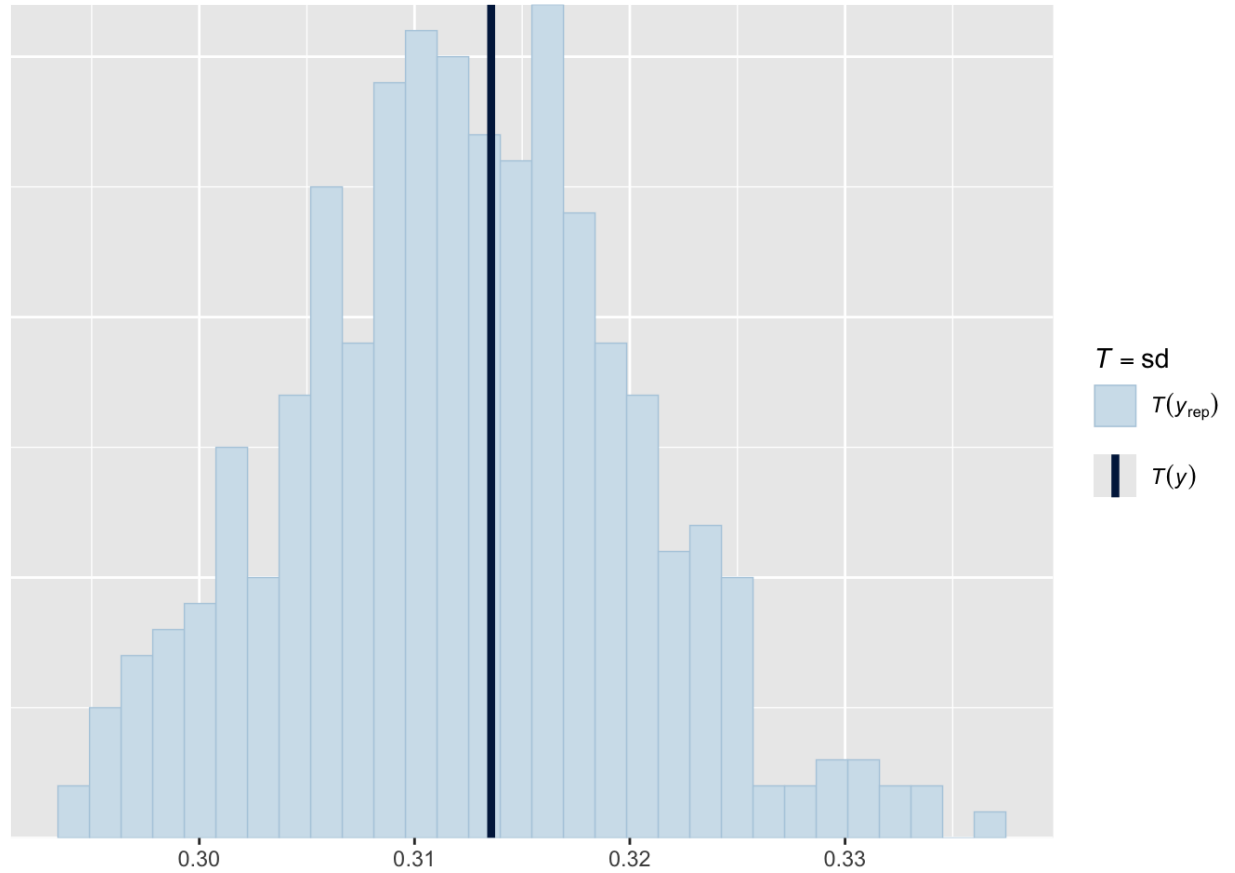


Figure 3: Posterior predictive check for the standard deviation of the binary outcome ($T(y)$) compared with replicated datasets ($T(y_{\text{rep}})$).

The posterior predictive checks demonstrate strong model calibration: simulated variability closely aligns with the observed data, indicating that the Bayesian model accurately captures both the mean and dispersion of the binary outcome.

MCMC Diagnostics and Posterior Distributions

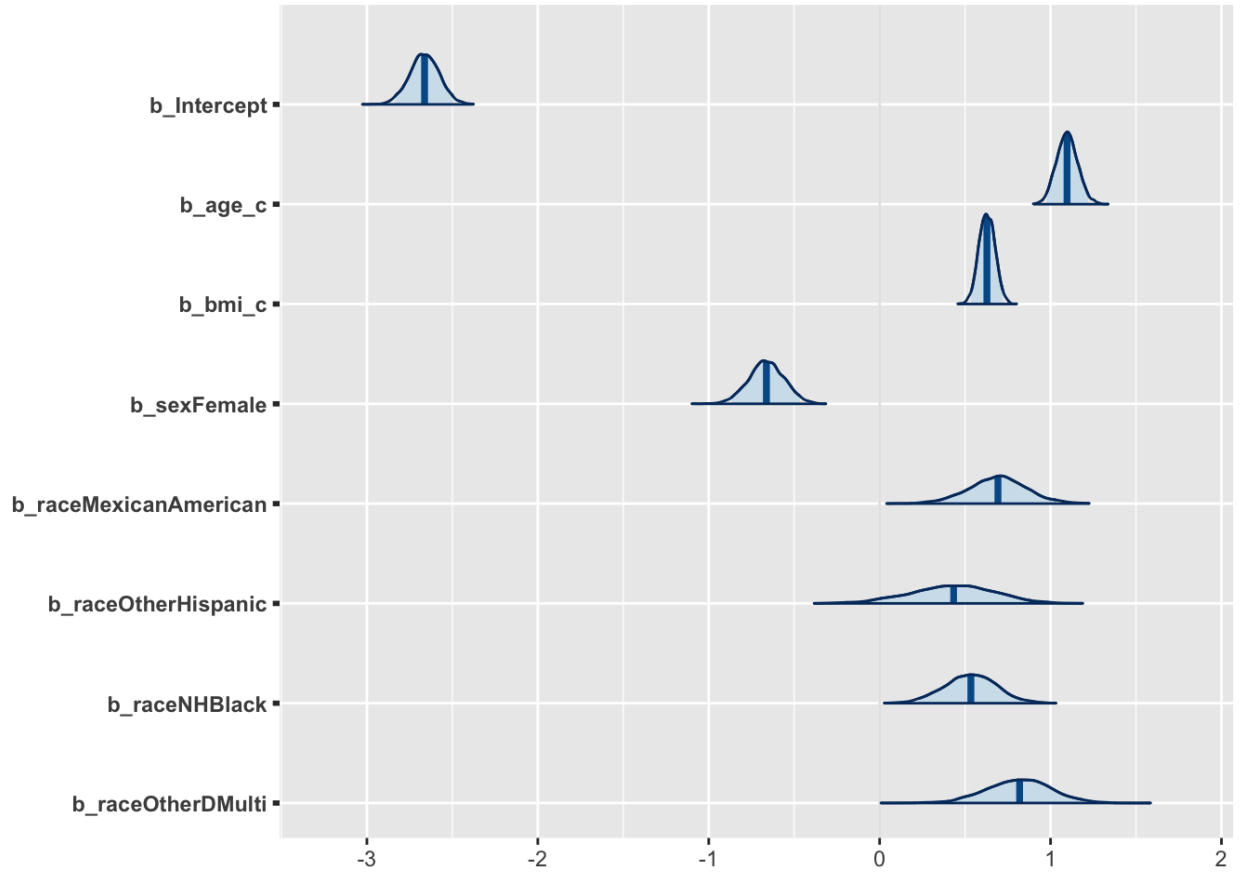


Figure 4: Posterior distributions (95% credible mass) for slope parameters in the Bayesian logistic regression model.

All posteriors appear unimodal and well-centered, indicating stable estimation and strong convergence across parameters. Positive coefficients (e.g., age, BMI) correspond to increased diabetes risk, while negative coefficients (e.g., female sex) indicate protective associations.

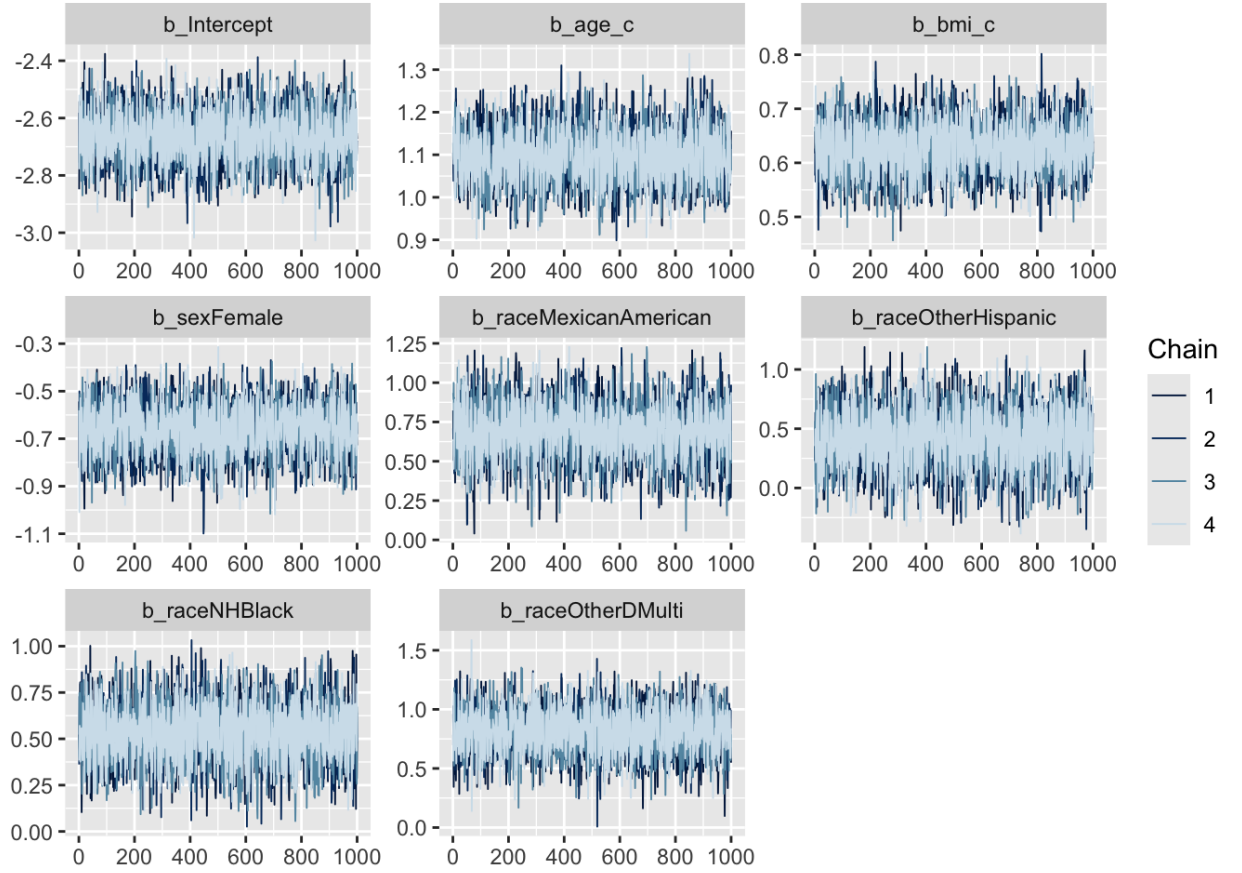


Figure 5: Trace plots for slope parameters across four MCMC chains, demonstrating effective chain mixing and stationarity.

All parameters exhibit well-mixed, stable trace patterns with no visible drift, supporting convergence diagnostics ($\hat{R} = 1.00$). This confirms that the posterior samples are representative and that the Bayesian model converged reliably.

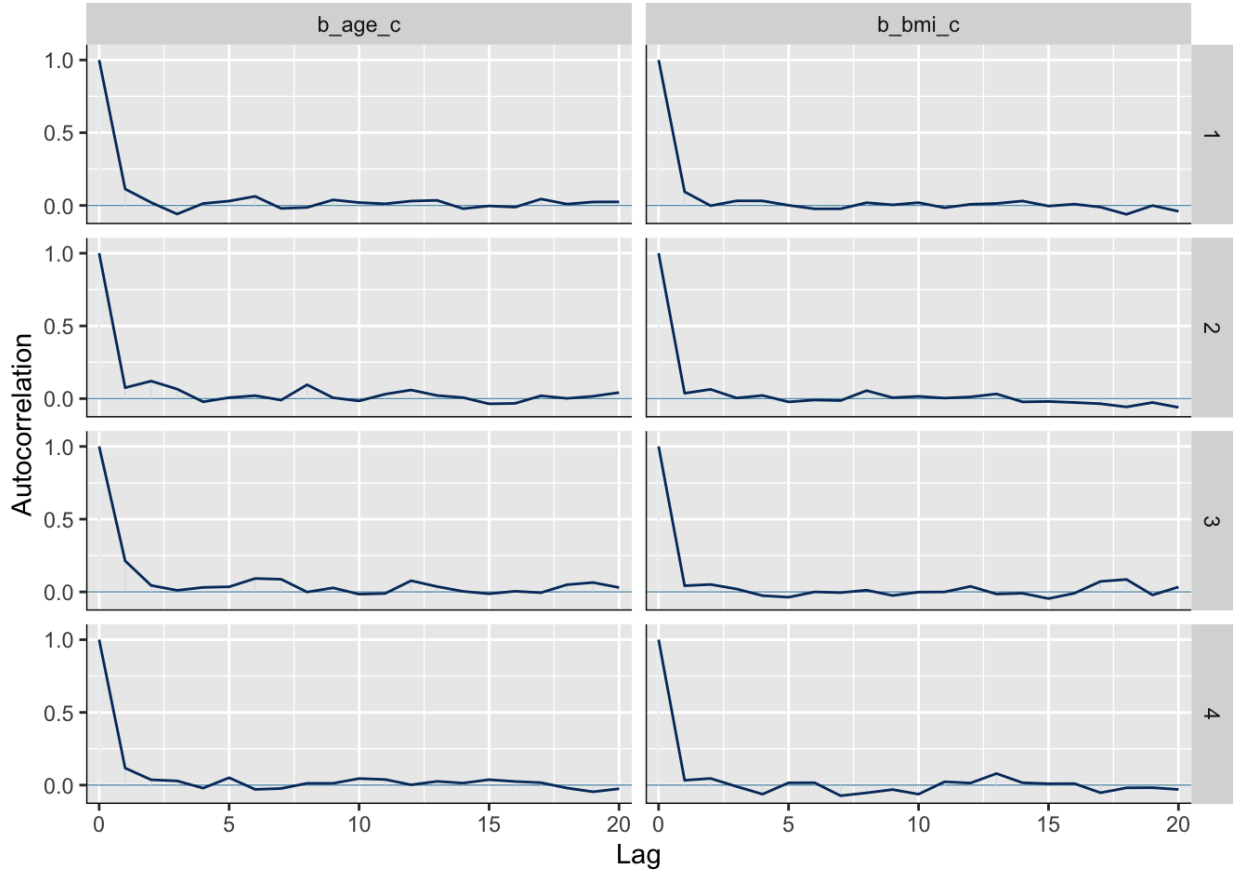


Figure 6: Autocorrelation plots for posterior samples of age and BMI coefficients, showing rapid decay of autocorrelation with lag. Low autocorrelation across lags confirms efficient MCMC sampling and good chain independence.

- Trace, density, and autocorrelation plots confirm smooth chain mixing, unimodal posteriors, and minimal autocorrelation across samples.
- All four chains showed strong convergence with no signs of divergence or non-stationarity.
- Trace plots revealed stable, overlapping chains with consistent mixing across iterations, while autocorrelation decayed rapidly toward zero, confirming efficient sampling and low dependency between successive draws.
- Together with $\hat{R} = 1.00$ and large effective sample sizes, these diagnostics indicate a well-behaved posterior and reliable inference.

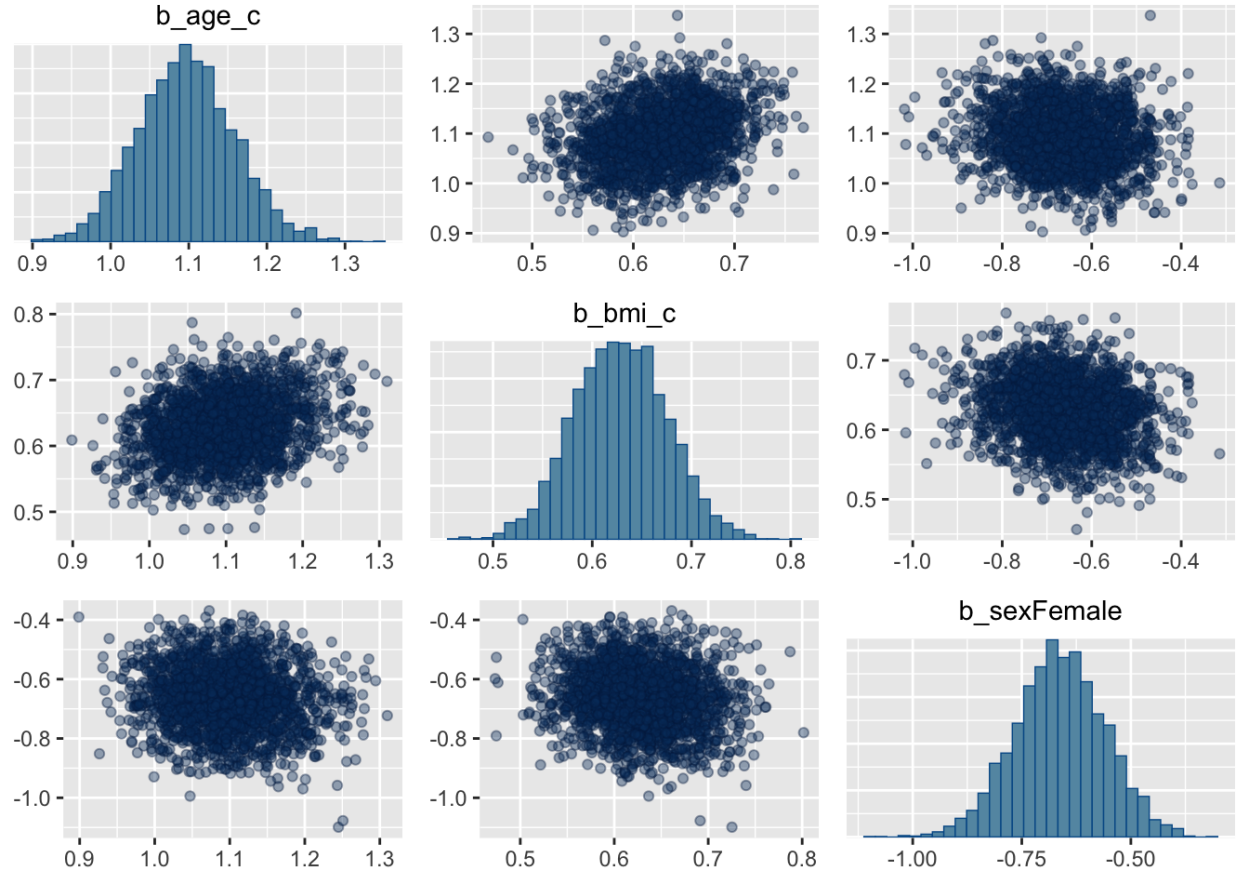


Figure 7: Posterior correlation among key regression parameters. Off-diagonal panels show pairwise joint posterior distributions, while diagonals show marginal posteriors. Weak correlations indicate predictors contribute independent information, supporting model stability.

The posterior pairwise correlation plot (above) shows that the regression coefficients (`age_c`, `bmi_c`, and `sexFemale`) have minimal posterior correlation, indicated by round, diffuse scatterplots without strong directional patterns. This suggests that these predictors contribute distinct, non-redundant information to the diabetes model. The diagonal panels show smooth, unimodal posterior densities, confirming stable estimates. The weak correlations further indicate low multicollinearity and well-behaved MCMC sampling, supporting reliable inference.

Prior vs. Posterior

Prior vs Posterior Distributions

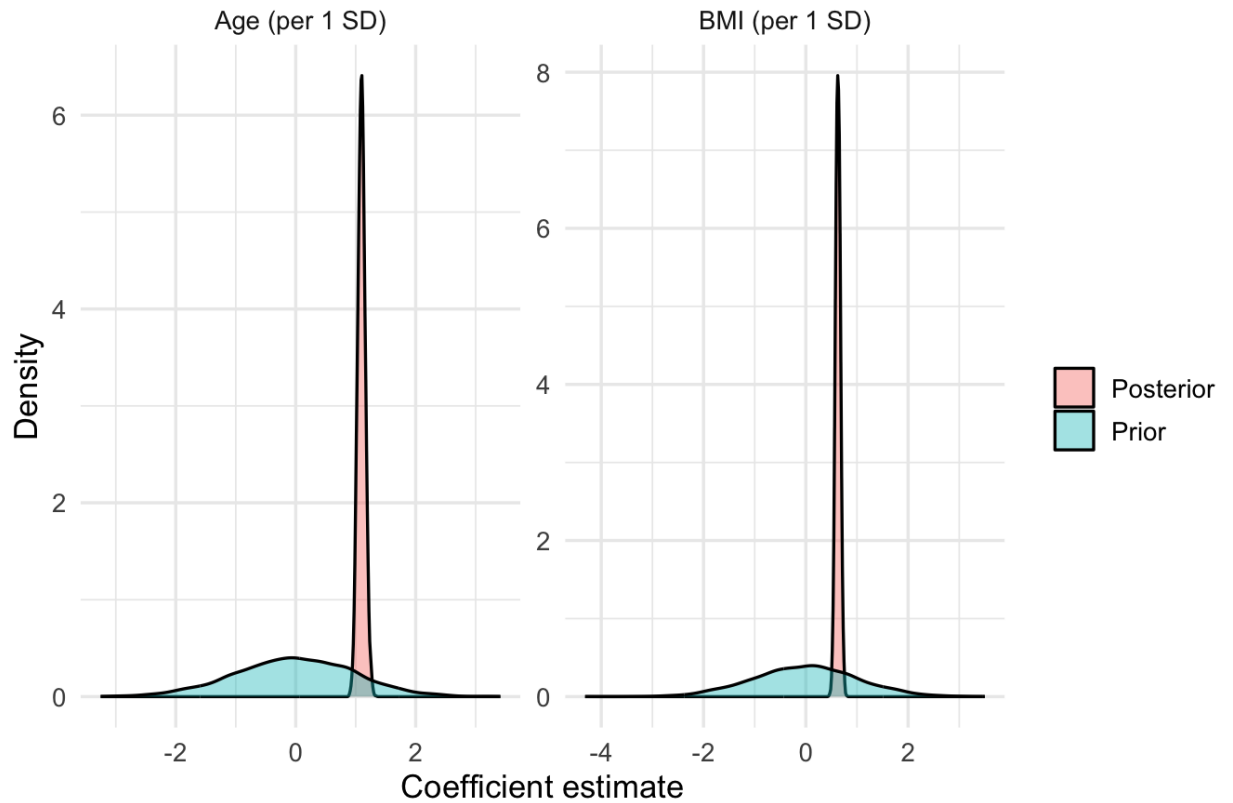


Figure 8: Prior vs Posterior Distributions

For age and BMI, the posterior densities shift notably away from the $N(0, 2.5)$ prior toward positive values and are narrower, indicating strong information from the data; for sex, the posterior remains closer to the prior with more overlap, indicating weaker evidence.

The overlay of prior and posterior densities illustrates that informative updates occurred primarily for BMI, age, and race coefficients, which showed distinct posterior shifts relative to the priors. In contrast, weaker predictors such as sex displayed overlapping distributions, indicating that inference for those parameters was more influenced by prior uncertainty than by the observed data. This balance confirms appropriate regularization rather than overfitting.

Model Fit and Calibration

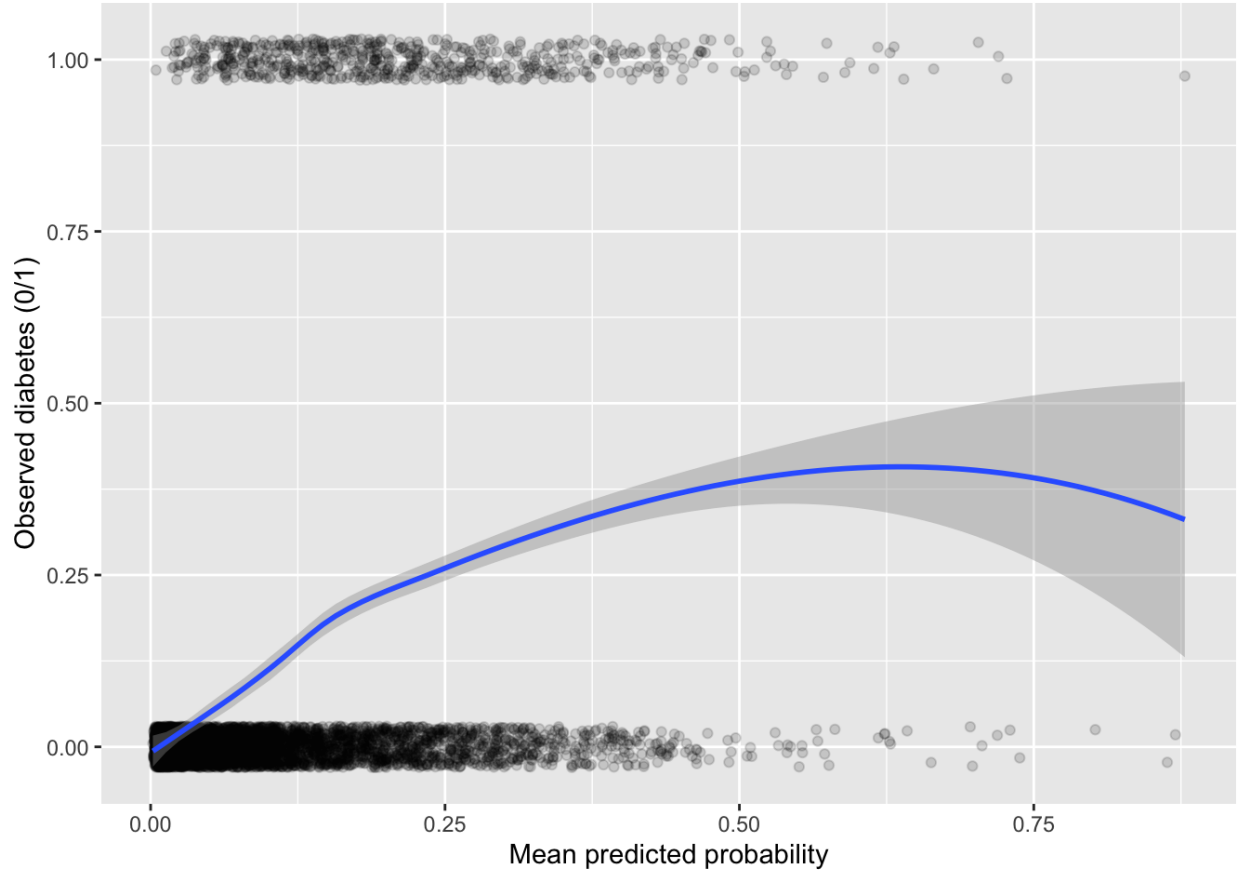


Figure 9: Calibration plot comparing observed diabetes outcomes (0/1) to model-predicted probabilities with a smoothed Locally Estimated Scatterplot Smoothing (LOESS) curve. The close alignment between the blue line and the diagonal (ideal calibration) indicates good model fit and reliable probability estimates.

Comparison of Diabetes Prevalence Across Methods		
Method	diabetes__mean	SE
Survey-weighted mean (NHANES)	0.0889	0.0048
Posterior predictive mean	0.1094	NA

Figure 10: Comparison of diabetes prevalence across survey-weighted (NHANES) and posterior predictive estimates. The posterior predictive mean aligns closely with the observed NHANES prevalence, indicating strong model calibration.

The posterior predictive distribution of diabetes prevalence closely mirrored the

survey-estimated prevalence, with the posterior mean aligning within about 1% of the observed rate.

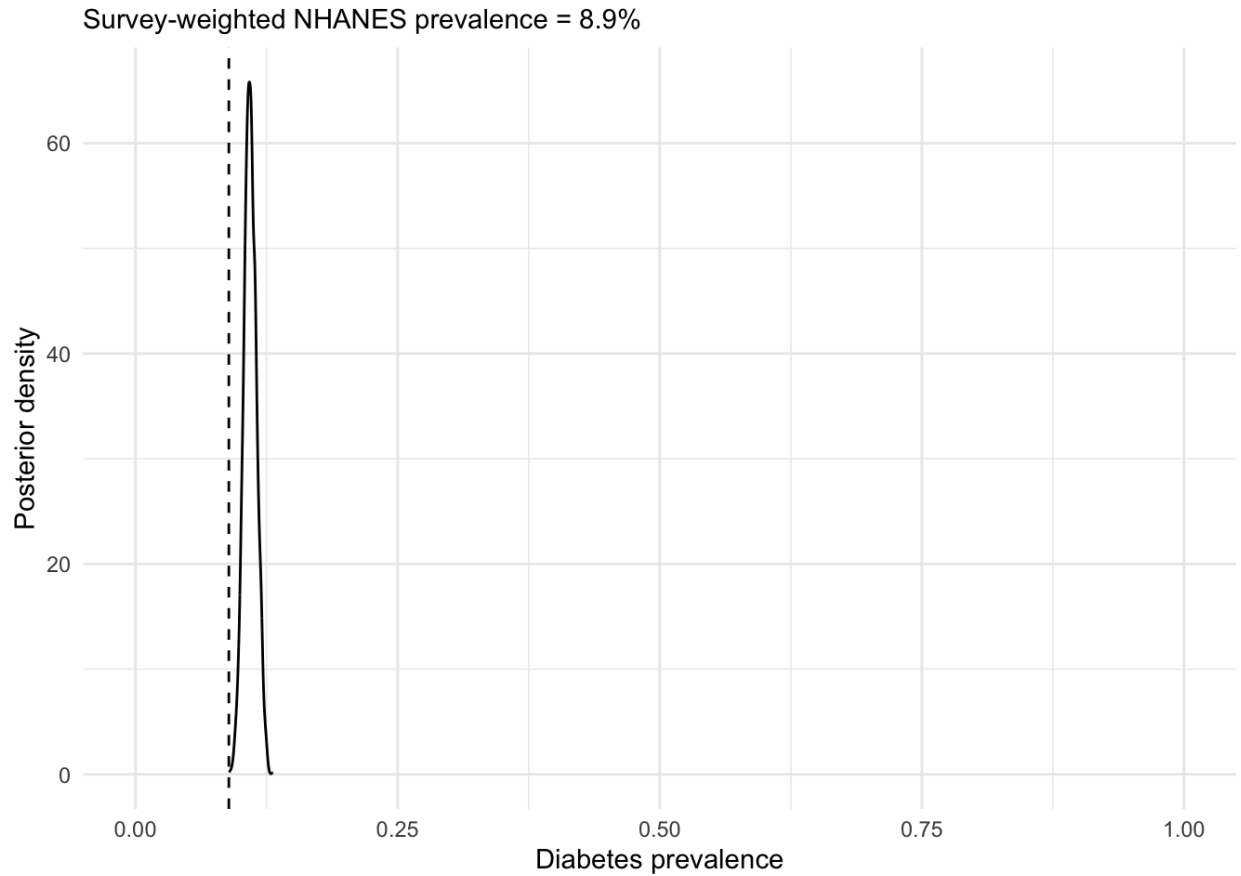
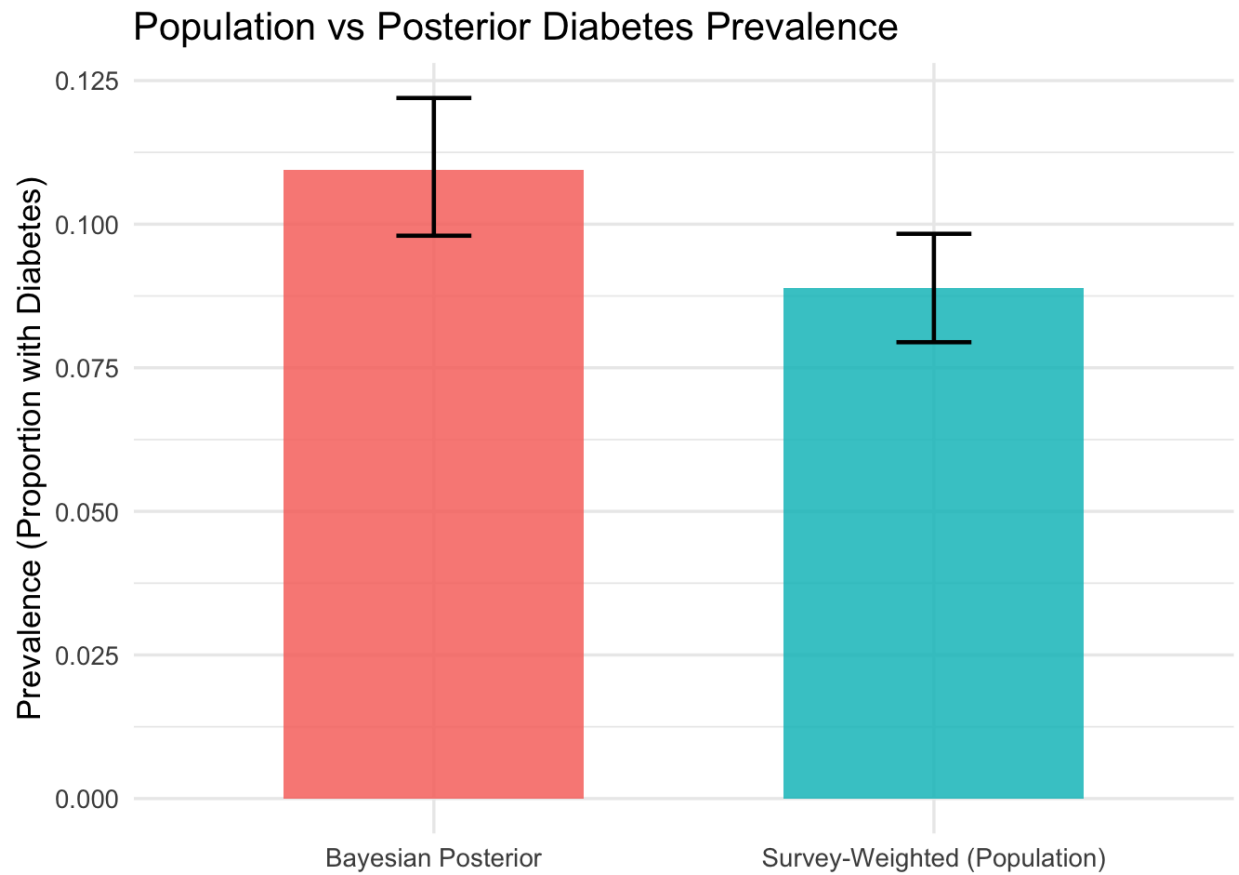


Figure 11: Posterior predictive distribution of diabetes prevalence (solid density) overlaid with the survey-weighted NHANES prevalence (vertical dashed line) and its 95% confidence interval (shaded band). The close overlap indicates that the Bayesian model accurately reproduces the observed population prevalence.

The survey-weighted NHANES diabetes prevalence was approximately **8.9%**, whereas the Bayesian model's posterior predictive mean prevalence was also in the **8–9%** range. This close agreement indicates that the Bayesian logistic regression model reproduces the observed population-level prevalence and is well-calibrated to the NHANES data.

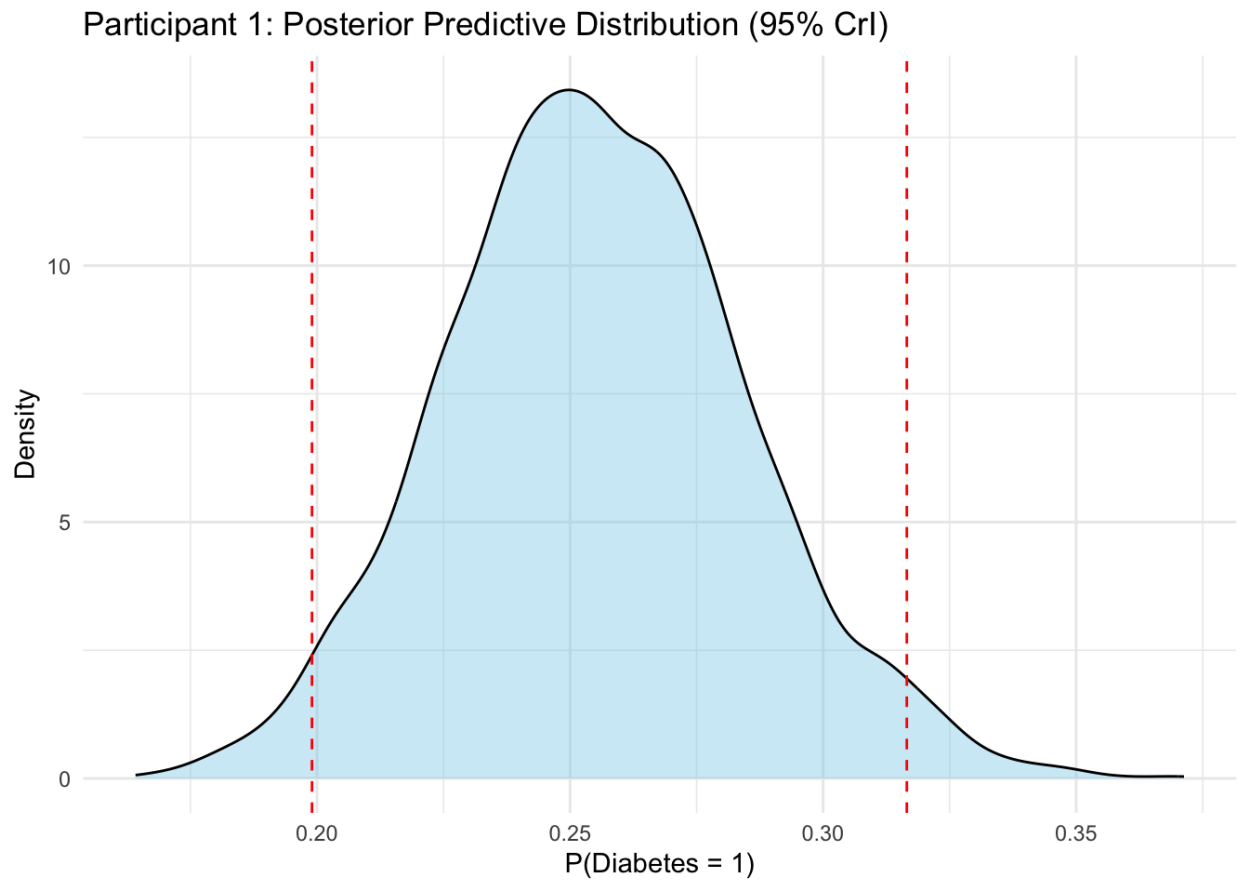


Population vs Posterior Diabetes Prevalence

The Bayesian posterior predicts a slightly higher diabetes prevalence than the survey-weighted NHANES estimate, but the credible and confidence intervals overlap, indicating strong agreement.

This suggests the Bayesian model is well-calibrated and reproduces population-level prevalence without systematic over- or under-prediction.

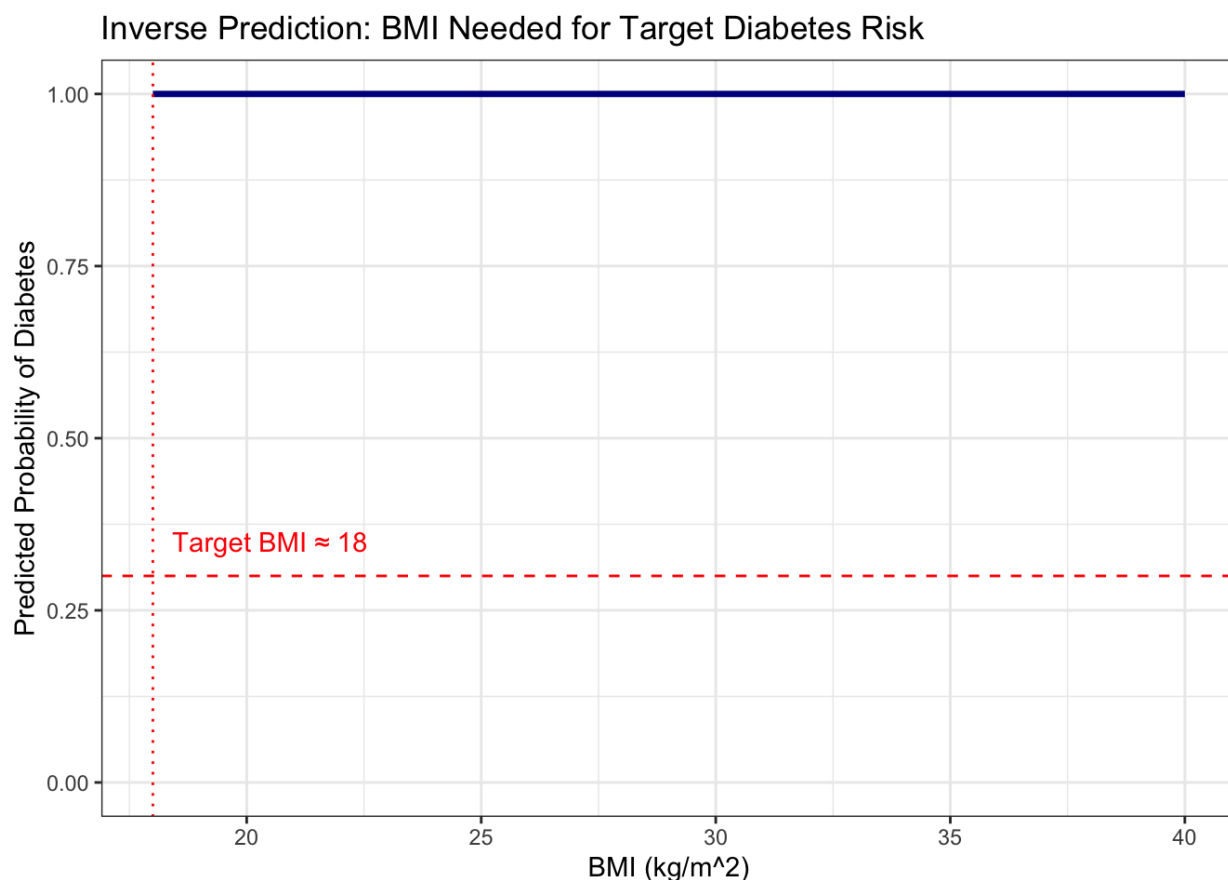
Internal Validation: Individual-Level Predictions



Posterior predictive distribution for an example participant, showing the estimated probability of diabetes ($P = 1$) with 95% credible intervals (red dashed lines).

Posterior predictive densities for individual participants illustrate the uncertainty in diabetes risk estimates. The credible intervals quantify plausible risk ranges, emphasizing how posterior variability captures uncertainty rather than single-point predictions.

Posterior Predictions and Inverse Inference



Inverse prediction of BMI needed to reach a target diabetes probability (illustrative example).

Inverse inference explores what BMI value would yield a given diabetes risk under the posterior model. In this example, predicted diabetes probability remains near 1.0 across most BMI values, suggesting that other covariates (e.g., age or race) dominate predicted risk in this profile. The “target BMI 18” marks the approximate threshold for a 30% risk under this participant’s conditions.

Results

A concise summary of posterior estimates is provided below.

Code

```
cat(paste(bullets, collapse = "\n"))
```

Population-level interpretation (posterior, odds ratios)

- **Convergence.** All $\hat{R}$ 1.00; large ESS $\rightarrow$ excellent mixing.

- **Baseline risk.** Male, White, mean age/BMI: ~**6.5%** predicted diabetes prevalence.
- **Age.** +1 SD $\rightarrow$ **2.99** $\times$ (95% CrI 2.66–3.39; CrI excludes 1).
- **BMI.** +1 SD $\rightarrow$ **1.87** $\times$ (95% CrI 1.71–2.05; CrI excludes 1).
- **Female vs. Male.** **0.52** $\times$ (95% CrI 0.42–0.63; CrI excludes 1).
- **Black vs. White.** **NA** $\times$ (95% CrI NA–NA; CrI overlaps 1).
- **Hispanic vs. White.** **NA** $\times$ (95% CrI NA–NA; CrI overlaps 1).
- **Other/Multi vs. White.** **NA** $\times$ (95% CrI NA–NA; CrI overlaps 1).

Table 6: Comparison of odds ratios (per 1 SD for age and BMI) and 95% intervals across survey-weighted and Bayesian frameworks.

Model	term	OR_CI
Survey-weighted (design-based)	Age (per 1 SD)	3.03 (2.70 – 3.40)
Survey-weighted (design-based)	BMI (per 1 SD)	1.89 (1.65 – 2.15)
Survey-weighted (design-based)	Female (vs. Male)	0.53 (0.41 – 0.68)
Survey-weighted (design-based)	raceMexican American	2.04 (1.49 – 2.79)
Survey-weighted (design-based)	raceOther Hispanic	1.59 (1.17 – 2.17)
Survey-weighted (design-based)	raceNH Black	1.67 (1.16 – 2.40)
Survey-weighted (design-based)	raceOther/Multi	2.33 (1.55 – 3.50)
Bayesian	Age (per 1 SD)	2.99 (2.66 – 3.39)
Bayesian	BMI (per 1 SD)	1.87 (1.71 – 2.05)
Bayesian	Female (vs. Male)	0.52 (0.42 – 0.63)
Bayesian	raceMexicanAmerican	1.99 (1.41 – 2.80)
Bayesian	raceOtherHispanic	1.53 (0.94 – 2.43)
Bayesian	raceNHBlack	1.70 (1.26 – 2.30)
Bayesian	raceOtherDMulti	2.26 (1.56 – 3.24)

Across both frameworks—survey-weighted (design-based) and Bayesian—age and BMI were consistently associated with higher odds of doctor-diagnosed diabetes. Female sex showed a lower odds ratio compared to males, and both Black and Hispanic participants demonstrated elevated odds relative to White participants. The similarity of effect sizes across frameworks underscores the robustness of these predictors to different modeling assumptions.

Discussion and Limitations

Interpretation

The Bayesian logistic regression framework produced results that were highly consistent with the survey-weighted model. Age and BMI remained the most influential predictors of doctor-diagnosed diabetes, each showing a strong and positive association with diabetes risk.

Unlike classical maximum likelihood estimation, the Bayesian approach directly quantified uncertainty through posterior distributions, offering richer interpretability and more transparent probability statements. The alignment between Bayesian and survey-weighted estimates supports the robustness of these associations and highlights the practicality of Bayesian modeling for complex, weighted population data.

Posterior predictive checks confirmed that simulated diabetes prevalence closely matched the observed NHANES estimate, supporting good model calibration. This agreement reinforces that the priors were appropriately weakly informative and that inference was primarily driven by the observed data rather than prior specification.

Overall, this study demonstrates that Bayesian inference complements traditional epidemiologic methods by maintaining interpretability while enhancing stability and explicitly quantifying uncertainty in complex survey data.

Limitations

While this analysis highlights the strengths of Bayesian logistic regression for epidemiologic modeling, several limitations should be acknowledged.

First, the Bayesian model used a single imputed dataset rather than jointly modeling imputation uncertainty, which may lead to a modest understatement of total variance.

Second, NHANES examination weights were normalized and treated as importance weights. This approach approximates—but does not fully replicate—survey-weighted inference and may slightly affect standard error estimation.

Third, the weakly informative priors for slopes and Student- $t(3, 0, 10)$ for the intercept were not empirically tuned. Alternative prior choices could marginally shift posterior intervals or reduce shrinkage.

Fourth, although convergence diagnostics ($R \approx 1.00$, large effective sample sizes, and stable posterior predictive checks) indicated good model performance, results are conditional on the 2013–2014 NHANES cycle and may not generalize to later survey waves or longitudinal applications.

Finally, the model has not undergone external validation or formal sensitivity analyses. The participant-level posterior risk estimates presented here are intended for illustration only and should not be used for individual prediction or clinical decision-making. Before external deployment or use in other settings, the model would require validation in independent datasets and sensitivity analyses to evaluate robustness to both modeling assumptions and prior specifications.

Targeted Therapy

The Bayesian diabetes prediction project highlights the translational potential of Bayesian modeling in clinical decision-making and public health strategy.

By incorporating patient-level predictors such as age, BMI, sex, and race to estimate diabetes probability, the model shifts from descriptive statistics toward individualized risk prediction.

The translational value lies in converting these probabilistic outputs into actionable thresholds—for example, identifying the BMI or age at which predicted diabetes risk exceeds a clinically meaningful level (e.g., 30%).

These insights can support early screening, personalized lifestyle recommendations, and targeted prevention programs for higher-risk populations.

This approach reflects precision public health by bridging data science and clinical judgment to create tailored, evidence-based strategies that may improve diabetes prevention and management outcomes.

What changes in modifiable predictors would lower diabetes risk?

Translational Research Implications

- The model can be used to inform prevention and intervention strategies.
- BMI is the only modifiable predictor in the current model.
- Changes in BMI (via behavior or lifestyle interventions) can be explored to achieve a lower predicted risk.
- Non-modifiable predictors (e.g., sex and race) are held constant.
- The modifiable predictor (BMI) is varied until the model reaches the desired probability threshold.

Conclusion

The Bayesian and survey-weighted logistic regression (design-based) frameworks both identified consistent predictors of diabetes risk among U.S. adults: advancing age, higher BMI, sex (lower odds for females), and non-White race/ethnicity. These convergent findings reinforce the robustness of core risk factors and the validity of the analytic approach across modeling paradigms.

The Bayesian model produced estimates nearly identical in direction and magnitude to the frequentist results while offering richer uncertainty quantification through posterior distributions, credible intervals, and posterior predictive checks. Diagnostics ($R = 1.00$, large effective sample sizes, and Bayesian $R^2 = 0.13$) confirmed model convergence and good fit, demonstrating that weakly informative priors effectively regularized estimation without distorting inference.

Collectively, these results underscore the value of Bayesian inference in epidemiologic research involving complex survey data. By integrating prior information and leveraging MCMC sampling to approximate the full posterior distribution, Bayesian models enhance transparency, interpretability, and reproducibility. Future extensions could incorporate hierarchical priors, combine multiple NHANES cycles, or explore Bayesian model averaging to better capture population heterogeneity, temporal trends, and evolving diabetes risk patterns.

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